Optimal Dynamic Allocation: Simplicity through Information Design*

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Abstract

We study dynamic nonmonetary markets where objects are allocated to unit-demand agents with private types. An agent's value for an object is supermodular in her type and the quality of the object, and her payoff is quasilinear in her waiting cost. We identify the welfare-maximizing mechanism in the class of direct-revelation mechanisms that elicit agents' types and assign them to objects over time. When the social planner can design the information disclosed to the agents about the objects, this mechanism can be implemented by a first-come first-served wait-list with deferrals. The optimal disclosure policy pools adjacent object types. Moreover, the hazard rate of the distribution of the agents' types determines the structure of the optimal disclosure policy, when the agents' utility function is separable. A single-peaked (single-dipped) hazard rate leads to the optimality of a lower censorship (upper censorship) policy.

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1 Introduction

This paper considers a dynamic assignment problem in a marketplace without money, where objects and agents, who have privately known preferences, arrive over time. A notable example is the allocation of organs from deceased donors for transplantation. Such organs are allocated through priority-based wait-lists. The median waiting time for a cadaver kidney in the U.S. is 3.6 years;[1] at the same time about 20% of the cadaver kidneys offered for transplantation are discarded.[2]

Wait-lists offer a natural way of rationing scarce resources, are simple to implement, and do not elicit much information. For example, patients decide only upon receiving an organ offer whether to accept it or not.[3] It is natural to ask whether there is any loss from using wait-lists (instead of more sophisticated mechanisms that elicit refined preferences) to allocate such scarce resources over time. This paper focuses on the allocation of resources that have a single-dimensional quality to agents with heterogeneous tastes for quality.

Our model features agents and objects that arrive over time at possibly different constant rates. Objects vary vertically in a single dimension of quality. Agents have single-dimensional private types that capture their preference for quality: an agent's value for an object is a supermodular function of the agent's type and the object's quality. An agent's payoff is her value for the (expected) object quality that she allocates minus her *waiting cost*, i.e., the cost she incurs while waiting for an object. Focusing on a single dimension captures the fact that the expected life span after a transplant, while depending on recipient characteristics, is positively correlated with the quality of the kidney [Wolfe et al., 2008].[4][5]

The social planner's objective is to maximize *social welfare*, defined as the agents' average payoff. Due to the supermodularity of the utility function, a positive assortative assignment maximizes the agents' average utility, but not necessarily their social welfare (which accounts for waiting costs). Since the agents have private types, we take a mechanism design approach. The goal is to design a direct revelation mechanism that maximizes social welfare over the space of all *steady-state* direct revelation mechanisms.

Focusing on *large* markets allows us to analyze a market with multiple agent types and object types (i.e., object qualities) and optimize over the space of all direct revelation mechanisms. This is in contrast to previous studies in the dynamic matching literature that restrict attention to optimizing over special classes of mechanisms [Su and Zenios, 2006, Bloch and Cantala, 2017,

[1] See https://www.kidney.org/news/newsroom/factsheets/Organ-Donation-and-Transplantation-Stats.

[2] Public housing is often allocated through wait-lists and lotteries. In the New York City Housing Authority has 173,762 public housing apartments with only 1% of these apartments available for occupancy, and with 181,090 families on the wait-list as of March 2019 (see https://www1.nyc.gov/assets/nycha/downloads/pdf/NYCHA-Fact-Sheet_2019_08-01.pdf).

[3] Though often some coarse preferences are revealed to the social planner when agents join these wait-lists.

[4] In the U.S. the quality of a kidney transplant is measured by the Kidney Donor Risk Index (see https://optn.transplant.hrsa.gov/resources/allocation-calculators/kdpi-calculator/).

[5] See, however, Agarwal et al. [2021] for an empirical treatment of organ allocation which also accounts for horizontal differentiation.

Leshno, 2017] or, alternatively, assume that the agents' types are observable [Baccara et al., 2020, Akbarpour et al., 2020b].

This paper makes two main contributions: characterizing the optimal direct revelation mechanism as a set of *monotone disjoint queues*, and implementing its outcome using a single wait-list with deferrals when the planner can design the information disclosed to agents about the objects.

Characterization of the optimal direct revelation mechanism

The optimal mechanism is characterized by a finite set of disjoint queues. Every agent, upon arrival, is assigned to a queue based on her type as reported. Within each queue, objects are allocated to agents according to their waiting times: when an object becomes available, it is assigned to the agent with the earliest arrival time present in the queue; the assignment is definitive and cannot be declined.

The objects are allocated to queues in a way such that, if two objects of distinct types are allocated to the same queue, then all objects with a type in between the two distinct types are also allocated to that queue. The same property holds for agents' types. The maximum number of queues required by the optimal mechanism is $2n + 1$, where n is the number of distinct object types.

We further elaborate the structure of optimal mechanisms in a common special case where the agent's utility is the product of her type and the object's quality. In this separable case, we show that if the hazard rate of the distribution of the agents' types is single-peaked (single-dipped) then the optimal mechanism *pools* a subset of the lowest (highest) object types into one queue, and sends every higher (lower) object type to a distinct queue.

The optimal mechanism has a natural indirect implementation, which maintains one queue for each of the types that is allocated to it. Agents who arrive observe waiting times and join a single queue upon arrival but cannot decline an offer. Waiting times depend on the number of queues and types in each queue. In the continuum model, waiting times are deterministic thus social welfare is maximized. In the discrete market model, fluctuations in waiting times will generate inefficiency. We find that the inefficiency vanishes quickly as the market grows large (Section 6).

In terms of methodology, we build our results upon the work of Kleiner et al. [2020], who characterize the extreme points of a set of functions that are majorized by a reference function. We first transform the planner's problem into maximizing *virtual payoff* by choosing an interim allocation rule that satisfies monotonicity and a certain majorization condition that ensures feasibility. If the virtual payoff function is increasing, the optimal mechanism allocates objects in a greedy manner, creating one separate queue for each object type. Otherwise, the optimal mechanism *pools* together some of the adjacent object types, creating one queue for each interval of pooled object types. To characterize the optimal mechanism in the latter case, standard ironing techniques in mechanism design [Myerson, 1981, Toikka, 2011] are not directly applicable, due to the concurrence

of two complicating factors: the supermodularity of the agents' values for objects and the capacity constraints of the heterogeneous object types.[6]

We take a different approach by applying Bauer's Maximum Principle [Bauer, 1958], implying that the virtual payoff-maximizing allocation rule attains its maximum at an extreme point of the set of allocation rules that satisfy monotonicity and a certain majorization condition that ensures feasibility. We then use the extreme point characterization result of Kleiner et al. [2020] to conclude that the interim allocation that maximizes virtual payoff has an *interval structure*, which is shown to correspond to a monotone disjoint queue mechanism.

Implementation through first-come, first-served (FCFS) wait-list with deferrals using information design

One feature of the optimal mechanism characterized above is that agents cannot decline the object they are assigned to. In numerous applications, however, agents wait in a *single* wait-list and can *decline* an assignment while keeping their position on the list. We ask whether offering this flexibility to agents can result in efficiency loss. We show that the answer to this question is negative if the planner can design the information disclosed to agents about the objects.

When objects' types are fully disclosed, higher-type objects are assigned to higher-type agents who value object quality more and, hence, are willing to wait longer for higher-quality objects. Therefore, waiting costs act as prices to clear the market, with higher-quality objects having higher prices. However, waiting costs are also sunk costs and therefore FCFS with full disclosure may not maximize social welfare. An information disclosure policy that does not fully disclose the object types may allow for changing these prices such that efficiency gains from lower prices offset losses from the non-assortative assignment of objects to agents.

We find that FCFS with deferrals can implement the same outcome as the optimal direct revelation mechanism when paired with an appropriate information disclosure policy. Agents decide whether to join the wait-list and objects are offered on a FCFS basis to the agents, who can decline offers while keeping their position on the wait-list. The information disclosure policy uses the same interval structure that is used in the optimal direct revelation mechanism; so, upon receiving an offer, an agent can only observe the interval containing the quality of the offered object. We show that the implemented outcome coincides with that of the optimal direct revelation mechanism.

The challenge in analyzing a single wait-list is accounting for the agents' strategic behavior who have to reason about the decisions of the other agents ahead of them on the list. We show that this dynamic game has a unique equilibrium when the information disclosure policy pools adjacent object types. In addition, there is such a disclosure policy under which the welfare at this equilibrium equals the welfare under the optimal disjoint queue mechanism.

[6] Remark 4.3 discusses in more detail the complications that arise in applying classic ironing approaches.

1.1 Related literature

This paper relates to several strands of literature. The first strand is the literature on dynamic matching and allocation of resources through queuing mechanisms.[7] Most of the papers in this area either focus on optimizing over special classes of mechanisms [Su and Zenios, 2006, Bloch and Cantala, 2017, Leshno, 2017], or assume that the agents' types are observable and optimize over general classes of mechanisms [Baccara et al., 2020, Akbarpour et al., 2020b].

A closely related work is Su and Zenios [2006] who consider a model with a continuum of object types and a finite number of agent types. Agents have no waiting costs and depart at an identical, constant rate. They seek to maximize the sum of agents' utilities over a special class of disjoint queue mechanisms whereas we establish that the optimal mechanism is a monotone disjoint queue mechanism. Su and Zenios [2006] further impose the following assumption to ensure optimality: Consider a large static market. If all agents and objects changed their type to the highest type and then objects were distributed among agents, the sum of the utilities would be smaller than if every agent changed her type to the lowest type and received an object of the lowest type.

Optimal mechanisms have also been identified within special classes of queuing mechanisms in settings that do not require pooling or information design [Baccara et al., 2020, Leshno, 2017]. Leshno [2017] focuses on allocative efficiency in an overloaded system and does not include waiting times in the efficiency objective,[8] and Baccara et al. [2020] consider a model of observable preferences that exhibits supermodularity.[9]

Unlike the work discussed above, Che and Tercieux [2020] consider a setting with homogeneous objects where the designer adjusts the agents' incentives for remaining in the queue through information disclosure about the queue length. They find that FCFS with no information disclosure is optimal in a rich domain.

Another related paper is Arnosti and Shi [2020], which studies a discrete-time continuum model for affordable housing allocation, in which agents vary in their need and further have independent preferences over identical developments.[10] The paper studies the trade-offs between matching and targeting and establish equivalences between certain common procedures such as independent and common lotteries and wait-lists with or without choice.

In contrast, we consider a general class of mechanisms in a setting with private information and optimize over the incentive-compatible mechanisms in that class. Another key contribution of our paper is that the optimal mechanism can be implemented by a FCFS allocation policy with a single queue for all agents coupled with an information disclosure policy that pools adjacent object

[7] The concept of negative externalities from waiting goes back to Naor [1969] and Vickrey [1969], who suggest using prices and tolls to improve efficiency.

[8] Leshno [2017] finds that, to minimize mismatches, it is optimal to use a buffer queue that encourages agents to wait for their desired object.

[9] Several papers consider settings in which preferences are dichotomous and determined by agents' compatibilities [Ashlagi et al., 2019, Akbarpour et al., 2020b].

[10] See also Thakral [2019].

types; we establish this result in Section 5.

Other related work concerns costly signaling ("money burning") settings, where mechanism design problems with the objective of maximizing welfare have been studied [Hartline and Roughgarden, 2008, Condorelli, 2012]. Hartline and Roughgarden [2008] study how to optimally allocate identical objects to maximize the residual surplus. Condorelli [2012] extends this to a setting with objects of different qualities. Both papers are concerned with the case of separable utilities. As noted by Condorelli [2012], his results are applicable to waiting-line auctions with identical objects. Our results contribute to the design of dynamic wait-list mechanisms with heterogeneous types of agents and objects. Moreover, our assumption of supermodular utilities requires novel methodologies compared to these two papers, as discussed previously. Similar to Condorelli [2012], Kleiner et al. [2020] study a static matching model with separable utilities as an application of their extreme-point characterization results; applying those results, they derive conditions under which the optimal mechanism is positive assortative or a lottery (i.e., pools all objects together).

In a concurrent work, Akbarpour et al. [2020a] use concavification techniques to study redistribution in a static market, where the objective is maximizing a linear combination of revenue and total surplus. They show that, under certain conditions, it is optimal to augment market mechanisms with lotteries that allocate some of the objects at a reduced price.

Theorem 5.5 connects information design to the design of simple allocation policies in matching markets. Several papers maximize social welfare via information design in a single queue that serves customers on a FCFS basis [Simhon et al., 2016, Lingenbrink and Iyer, 2017, Anunrojwong et al., 2020, Che and Tercieux, 2020]. These papers identify optimal signaling schemes for recommending agents whether to join the queue or take their outside option under certain preference structures. Anunrojwong et al. [2020] study an unobservable FCFS queue where agents may choose an outside option over joining the queue. Lingenbrink and Iyer [2017] study optimal information disclosure in an unobservable single-server queue where agents decide whether to join the queue, and find that a binary signaling mechanism is optimal.

2 Setup

Consider a market in which agents and objects arrive over time to a pool, each with a flow of a constant rate. The arrival rate of agents is normalized to 1. (Section 6 provides counterparts to our findings in discrete markets where the arrival processes are Poisson.) Every object has a type that belongs to a finite set $\Omega = \{\omega_0, \omega_1, \ldots, \omega_n\}$, where $\omega_0, \ldots, \omega_n$ are distinct nonnegative reals ordered increasingly. An object of type ω_i has a *higher* type than an object with type ω_j if $\omega_i > \omega_j$. We also refer to an object type as its *quality*.

Objects of type $\omega \in \Omega$ arrive at the pool at a constant rate of $F(\omega)$. There is an *abundant* supply of objects of type ω_0: the arrival rate of these objects, $F(\omega_0)$, equals the arrival rate of agents, 1. One may interpret ω_0 as the outside option.

Let $\Omega_+ = \{\omega_1, \ldots, \omega_n\}$, and $N = \sum_{\omega \in \Omega_+} F(\omega)$. We assume that the market is *over-demanded*, i.e., $N < 1$.[11]

Every agent has a private type θ that is independently and identically distributed (i.i.d.) across all agents according to a continuous CDF G with PDF g and support $\Theta = [0, \bar{\theta})$, where $\bar{\theta} \in \mathbb{R}_+ \cup \infty$.[12] The utility that an agent derives from a random object depends on the agent's type and the object's expected quality. Formally, the utility that an agent of type θ assigns to a lottery over objects is $u(\theta, \overline{\omega})$, where $\overline{\omega}$ denotes the expected object type of the lottery and $u : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$. In particular, we make the following assumptions about the function u.

Assumption 2.1 (Spence-Mirrlees property)**.** *The function u is strictly supermodular: $\frac{\partial^2 u}{\partial \theta \partial \omega} > 0$.*

Assumption 2.2. *The function $\frac{\partial u}{\partial \theta}(\theta, \omega)$ is convex in ω.*

Assumption 2.3 (Smoothness)**.** *The utility function u is of class C^3.*

Assumption 2.4. *For all $\omega \in [\omega_0, \omega_n]$, $u(0, \omega) = 0$.*

A commonly considered special case satisfying these assumptions is the case of *separable* utilities $u(\theta, \omega) = \theta\omega$, where the agent would be an expected utility maximizer. (Appendix H studies another example where the agents' utilities are supermodular, but not separable, due to their heterogeneous *default options*.) The first assumption asserts that agents of higher types appreciate quality more. The second assumption is a technical assumption that is required for applying Bauer's Maximum Principle [Bauer, 1958]; it ensures that the problem of choosing an allocation rule that maximizes virtual payoff has an *extreme point* solution.[13] The third assumption is required in the characterization of [Rochet, 1987] for allocation rules that are truthfully implementable in dominant strategies. The last assumption ensures that the lowest type agent $\theta = 0$ derives no utility from being assigned any object.[14]

A *mechanism*, typically denoted by $\mathcal{M}$, elicits from every agent her type upon her arrival and assigns her an object at a time (possibly) after her arrival, as detailed below.

Every agent reports her type to the mechanism upon arrival to the pool. The mechanism then assigns an *allocation timeline* $\mathcal{P}_{\mathcal{M}}(\theta)$ to the agent based on her reported type θ. An allocation timeline $\mathcal{P}_{\mathcal{M}}(\theta)$, also denoted by $\mathcal{P}^{\theta}_{\mathcal{M}}$, is a joint probability distribution over object type and waiting time: for $\omega \in \Omega$, $\mathcal{P}^{\theta}_{\mathcal{M}}(t, \omega)$ is the probability that the agent is assigned an object with type ω at most t units of time after her report time. We focus on the set of mechanisms in which the pool remains at a *steady state*; that is, the density of agents of type θ and waiting time s present in

[11] Our results apply to under-demanded markets *mutatis mutandis*.

[12] In the case of a continuum of agents here, this is equivalent to assuming that the types of agents arriving at every time are distributed according to G.

[13] By Bauer's Maximum Principle, an upper semicontinuous convex real functional h over a compact convex subset K attains its maximum at an extreme point of K.

[14] This assumption significantly simplifies notation in the proof but is not crucial, and the proof carries over without it.

the pool at any time remains the same across time, for every $\theta \in \Theta$ and $s \geq 0$. Therefore, the allocation timeline assigned to an agent is assumed to be a function only of her reported type, and not the composition of the pool. (In Section 6 we consider a discrete setup where the steady state assumption is relaxed.) The function $\mathcal{P}_{\mathcal{M}}(\cdot)$ is called the mechanism's *allocation rule*.

The expected *waiting cost* of an agent who reports a type θ to a mechanism $\mathcal{M}$ is $\mathbb{E}_{t\sim\mathcal{P}^{\theta}_{\mathcal{M}}}[c(t)]$, where the random variable t is the waiting time of the agent under the allocation timeline $\mathcal{P}^{\theta}_{\mathcal{M}}$ and $c : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly increasing function that determines the agents' waiting costs.

The *payoff* $v_{\mathcal{M}}(\theta';\theta)$ of an agent of type θ from reporting type θ' to a mechanism $\mathcal{M}$ is quasi-linear in waiting costs: it equals the utility that the agent assigns to the allocation timeline $\mathcal{P}^{\theta'}_{\mathcal{M}}$ minus the expected waiting cost she incurs under $\mathcal{P}^{\theta'}_{\mathcal{M}}$.

A mechanism is *incentive-compatible* (IC) if $v_{\mathcal{M}}(\theta;\theta) \geq v_{\mathcal{M}}(\theta';\theta)$ for every $\theta, \theta' \in \Theta$; i.e., the payoff of every participating agent is maximized when she reports her true type to the mechanism. A mechanism is *individually rational* if the expected payoff of every participating agent is positive in that mechanism.

Remark 2.5. *An alternative definition for individual rationality is one in which an agent, at every time after her arrival, computes her expected present value (payoff) from remaining in the pool conditional on not being matched thus far, and immediately abandons the pool if the expected present value becomes nonpositive. We later discuss that the optimal mechanism that we find satisfies this stricter definition of individual rationality.*

The planner's objective is choosing an individually rational and incentive-compatible mechanism with the highest *social welfare*, as formally defined next. The *social welfare* generated by $\mathcal{M}$ is defined by $W_{\mathcal{M}} = \mathbb{E}_{\theta\sim G}[v_{\mathcal{M}}(\theta;\theta)]$. (We define social welfare as the simple average of the agents' payoffs for expositional simplicity. Appendix D.1 extends our results to the case of weighted average.) An individually rational and incentive-compatible mechanism $\mathcal{M}^*$ is *optimal* if there is no other such mechanism $\mathcal{M}$ such that $W_{\mathcal{M}^*} < W_{\mathcal{M}}$.

3 Motivating example

Before formally presenting our main results, we provide a simple example to give some intuition about the main economic forces in play. In this example the agents' utilities are separable, i.e., $u(\theta,\omega) = \theta\omega$. There are two object types $\Omega_+ = \{\omega_1, \omega_2\}$ with equal arrival rates $F(\omega_1) = F(\omega_2) = \frac{1}{4}$ (thus $N = \frac{1}{2}$). Also, G is the uniform distribution over the unit interval $\Theta = [0, 1]$.

We consider a class of queuing policies, which we define informally throughout this example. (The formal definitions and main results appear in the next section.) A policy in this class involves two queues q_1, q_2. When an object of type ω_1 arrives to the market, it is sent to q_2 with probability $\epsilon \in (0, 0.5]$ and to q_1 with probability $1 - \epsilon$. Symmetrically, an object of type ω_2 is sent to q_1 with

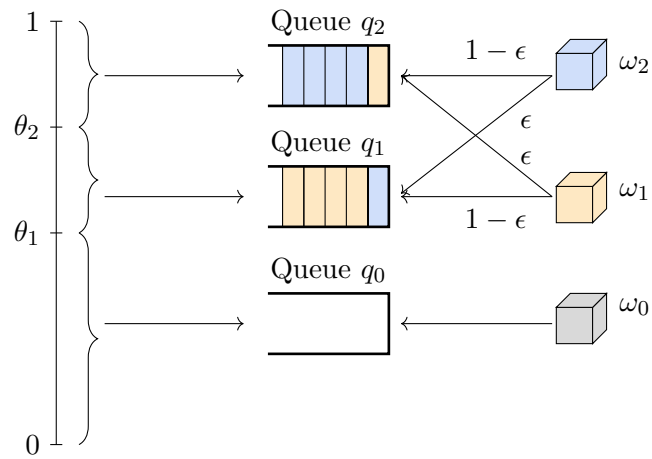


Figure 1: Illustration for the example in Section 3. Agents with types above θ_2 join queue q_2, agents with types between θ_1 and θ_2 join q_1 and agents with types below θ_1 do not join any queue and take their outside option. For $\epsilon \in (0, 0.5]$, each queue q_i, $i \in \{1, 2\}$, allocates a fraction $1-\epsilon$ of ω_i objects and a fraction ϵ of $\omega_{i'}$ objects, $i' \neq i$.

probability $\epsilon \geq 0$ and to q_2 with probability $1-\epsilon$. When an object is sent to a queue, it is allocated to the agent with the earliest arrival time in that queue and then the agent leaves immediately.

Every agent, upon her arrival, observes the waiting cost at each queue (i.e., observes the queue lengths) and chooses to enter the queue that maximizes her expected payoff, computed as her expected utility from allocating an object minus her expected waiting cost. Is there a *steady state* under this policy where the queue lengths do not change across time, and what would the waiting costs be there?

Let c_1, c_2 respectively denote the waiting costs for q_1, q_2 in such a steady state (Figure 1). There would be an *indifferent* agent type θ_2 who obtains the same payoff from joining either queue. Due the the supermodularity of u, all agents with type higher than θ_2 would join q_2. There also exists θ_1 such that every agent with a type lower than that would not join either queue, thus attaining a payoff of 0. The rest of the agents (those with types in between θ_1 and θ_2) would join q_1.

Since the arrival flow of agents to each queue should equate the arrival flow of objects to that queue, we must have $\theta_1 = \frac{1}{2}$ and $\theta_2 = \frac{3}{4}$. Then, to find the steady state waiting costs, we write the *indifference equation* for the agent type θ_1, equating her payoff from joining q_1 and not joining either queue:

$$\theta_1 \left((1-\epsilon)\omega_1 + \epsilon\omega_2\right) - c_1 = 0.$$

Similarly, we write the indifference equation for θ_2, equating her payoff from joining either of the queues:

$$\theta_2 \left((1-\epsilon)\omega_1 + \epsilon\omega_2\right) - c_1 = \theta_2 \left((1-\epsilon)\omega_2 + \epsilon\omega_1\right) - c_2.$$

Solving for c_1, c_2 from the indifference equations gives

$$c_1 = ((1-\epsilon)\omega_1 + \epsilon\omega_2)/2,\ c_2 = (\epsilon - 1/4)\,\omega_1 + (3/4 - \epsilon)\,\omega_2. \tag{3.1}$$

We then can compute the social welfare by a straightforward integration:

$$\int_{\theta_1}^{\theta_2} (\theta((1-\epsilon)\omega_1 + \epsilon\omega_2) - c_1)\,dt + \int_{\theta_2}^{1} (\theta(\epsilon\omega_1 + (1-\epsilon)\omega_2) - c_2)\,dt = \frac{1}{32}\left((3-2\epsilon)\omega_1 + (1+2\epsilon)\omega_2\right).$$

Denoting the right-hand side by $W(\epsilon)$, we observe that $W'(\epsilon) = \frac{\omega_2 - \omega_1}{16} > 0$. Incentive compatibility imposes $\epsilon \leq \frac{1}{2}$, as it requires the expected object type allocated by higher agent types to be higher. Over all $\epsilon \in [0, \frac{1}{2}]$, the choice of ϵ that maximizes social welfare is then $\epsilon = \frac{1}{2}$, i.e., when the two object types are completely pooled together.

The intuition is as follows. Pooling more (i.e., increasing ϵ) reduces the waiting cost at q_2, but increases the waiting cost at q_1. The waiting cost at q_2, however, decreases at a faster rate than the rate at which the waiting cost at q_1 increases. This can be seen from (3.1). Intuitively, the reason is that a higher-type agent has stronger incentives to wait more for an additional increment in quality than a lower-type agent does. The average waiting cost thus decreases with ϵ; in this example, to an extent that it offsets the allocative efficiency loss from pooling (i.e., from allocating objects to agents in a nonassortative manner).

Does the pooling solution $\epsilon = \frac{1}{2}$ correspond to an optimal mechanism? Putting the above example aside, what do optimal mechanisms generally look like? Can we construct the optimal mechanism? We address these questions in the next section.

4 Characterization of the optimal mechanism

To characterize optimal mechanisms generally in our setup, we first define a simple class of mechanisms that contains the optimal mechanism. The formal definition is presented after the following brief description. In a mechanism of this class, every agent is assigned to one of the finitely many queues in the mechanism based on her reported type. The agent waits in the queue until she is assigned an object; then, she immediately leaves the pool. The assignment of agents to the queues is such that, if two agents of types θ, θ' are assigned to the same queue, then every agent of a type in between θ, θ' is also assigned to that queue. Objects are sent to the queues such that, if two objects of distinct types ω, ω' are sent to the same queue, then every object with a type (strictly) in between ω, ω' is also sent to that queue. An object sent to a queue is assigned to the agent with the earliest arrival time present in the queue. The assignment is definitive and cannot be declined by the agent. We next formalize this definition.

Monotone disjoint queue mechanisms. A *monotone disjoint queue* mechanism is characterized

by a number k and a strictly increasing sequence of positive reals $\theta_0 = 0, \theta_1, \ldots, \theta_k, \theta_{k+1} = \overline{\theta}$. The mechanism features a finite number of queues $q_0, q_1, \ldots, q_k$. We say that q_j has a higher type than q_l if $j > l$. An agent who reports a type $\theta \in [\theta_i, \theta_{i+1})$ for $i \geq 0$ is assigned upon arrival to the queue q_i.

Every object immediately upon arrival is sent to one of the queues $q_0, \ldots, q_k$ to be offered to the agents in that queue. This is done to satisfy the following conditions: (i) for every object type ω and every queue q_i, objects of type ω are sent to q_i at a constant rate (possibly 0) over time, (ii) only objects of type ω_0 are sent to the queue q_0, and every agent assigned to q_0 is immediately allocated an object of type ω_0, (iii) the arrival rate of objects to the queue q_i is $G(\theta_{i+1}) - G(\theta_i)$ for all $i \geq 1$, and (iv) for any two objects of distinct types, the higher-type object is sent to a queue that has the same or a higher type than the type of the queue to which the lower-type object is sent.

Figure 2 illustrates the assignment of objects to queues (left panel) and the assignment of agents to queues (right panel) in a monotone disjoint mechanism. We note that these figures are in the same setting; hence, the colored areas in Figure 2 (right) correspond to the object types in Figure 2 (left).

Note that the class of incentive-compatible mechanisms is generally strictly larger than the class of monotone disjoint queue mechanisms. For example, the simple incentive-compatible mechanism we studied in Section 3 involves pooling of adjacent object types but is not a monotone disjoint queue mechanism since it violates condition (iv) in the definition above.

In a monotone disjoint queue mechanism, the pool remains at a steady state because the arrival rate of agents to each queue equals the arrival rate of objects to that queue (and agents cannot decline the objects that are assigned to them). Therefore, the length of each queue in such a mechanism remains constant across time. The next proposition characterizes the queue lengths in an incentive-compatible monotone disjoint queue mechanism.

Proposition 4.1. *Let $\overline{\omega}_i$ denote the average object type that is sent to the queue q_i in an incentive-compatible monotone disjoint queue mechanism. Then, the length of queue q_i is*

$$l_i = \begin{cases} c^{-1}(u(\theta_1, \overline{\omega}_1) - u(\theta_1, \omega_0)) & \text{for } i = 1, \\ c^{-1}(u(\theta_i, \overline{\omega}_i) - u(\theta_i, \overline{\omega}_{i-1}) + c(l_{i-1})) & \text{for } i > 1. \end{cases}$$

The above proposition thus uniquely characterizes the allocation rule associated with an incentive-compatible monotone disjoint queue mechanism.

Theorem 4.2. *The optimal mechanism, $\mathcal{M}^*$, is a monotone disjoint queue mechanism.*

Two remarks about the optimal mechanism are in order. First, we note that objects of each type can be sent to at most three distinct queues, which should be ordered consecutively. (Otherwise,

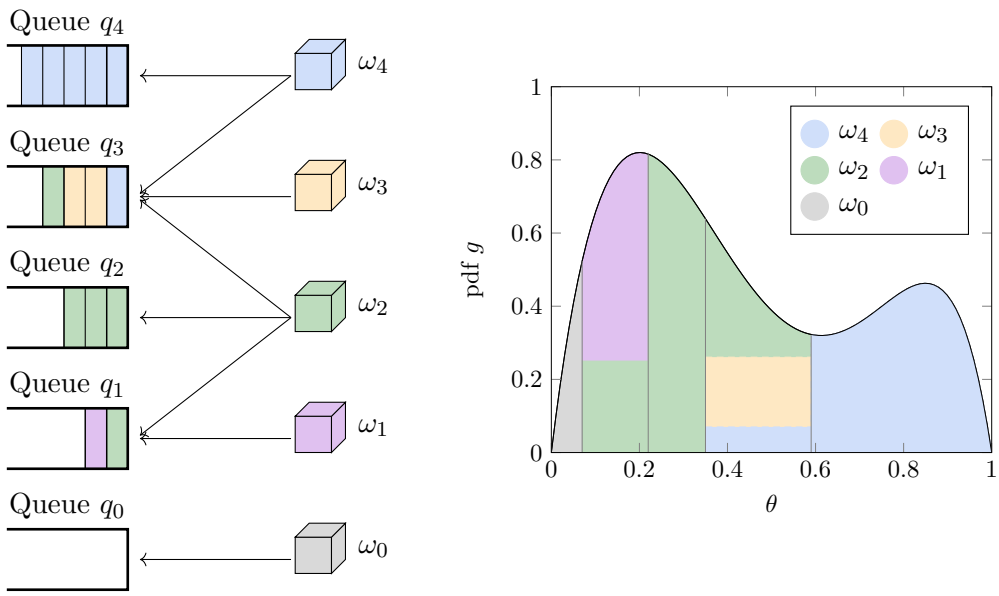


Figure 2: **Left**: The figure depicts an example of a monotone disjoint queue mechanism. Consider queue q_3; all objects of type ω_3 (yellow) are allocated, together with a fraction of objects of the adjacent types ω_2 (green) and ω_4 (blue), only to agents who join queue q_3. All agents that join queue q_3 have the same expected object type. Furthermore, note that the definition of monotone disjoint queue mechanisms (see condition (iv)) requires that for any two objects of distinct types, the higher-type object must be sent to a queue of the same or higher type than the type of the queue to which the lower-type object is sent. In this particular example, ω_4 is sent to q_4 and q_3, which implies that ω_3 can be sent at most to q_3. For similar reasons, an alternative mechanism, where objects of non-subsequent types ω_1, ω_3, and ω_4 are allocated through the same queue whereas all objects of type ω_2 (purple) are allocated through a different queue, does *not* belong to the class of monotone disjoint queue mechanisms. **Right**: Agents with a type below a certain threshold are assigned to the queue q_0 and receive objects of null type ω_0 (gray). This threshold, together with three other thresholds, partitions the area below the PDF into five areas. Each area corresponds to a queue in the left figure. Note that agents belonging to the same interval receive the same expected object type and experience the same expected waiting cost.

it is possible to merge two of the queues without changing the outcome for any agent.) This also implies that the number of queues used in the mechanism is at most $2n + 1$.

Second, the optimal mechanism satisfies the stronger notion of individual rationality mentioned in Remark 2.5. This holds because, as the agent spends more time in the queue, the expected waiting cost that she incurs decreases, but the expected object type that she will be allocated remains the same; i.e., the agent's *present value of waiting* increases as she spends more time in the queue.

We next describe the key steps in the proof of Theorem 4.2. The formal proof is in the appendix. After outlining the proof, we focus on the case of separable utilities, investigate the relation between the hazard rate of G and the set of object types that are pooled by the optimal mechanism, and

discuss the construction of disjoint queue mechanisms.

4.1 Proof approach

The proof is based on the notion of an *interim* allocation rule $X(\theta)$, which gives the interim expected object type assigned to an agent with reported type θ. The interim allocation rule of any mechanism must satisfy the set of interim incentive compatibility and individual rationality conditions for every agent type. We show that the solution to the planner's optimization problem with respect to the set of interim conditions indeed corresponds to a monotone disjoint queue mechanism, as outlined below.

We first recall some terminology from *majorization* theory (see, e.g., [Hardy et al., 1929, Kleiner et al., 2020]). For two nondecreasing interim allocation rules $X, Y \in L^1(\Theta)$,[15] we say that X *majorizes* Y, denoted by $Y \prec X$, if

$$\int_\theta^{\bar\theta} Y(s)\,\mathrm{d}G(s) \le \int_\theta^{\bar\theta} X(s)\,\mathrm{d}G(s) \text{ for all } \theta \in \Theta,$$
$$\int_0^{\bar\theta} Y(s)\,\mathrm{d}G(s) = \int_0^{\bar\theta} X(s)\,\mathrm{d}G(s).$$

To prove the theorem, we first transform the planner's problem into choosing an interim allocation rule X

$$\sup_{X\in\mathcal{S}} \int_0^{\bar\theta} \phi(\theta, X(\theta))\,\mathrm{d}G(\theta), \tag{4.1}$$

where

$$\phi(\theta, x) = \frac{1-G(\theta)}{g(\theta)} \cdot u_1(\theta, x)$$

is the *virtual payoff* function[16] and $\mathcal{S}$ is the set of all interim allocation rules of all incentive-compatible and individually rational mechanisms.

We then show that the virtual-payoff-maximizing allocation rule in (4.1) belongs to a subset of allocation rules

$$\mathcal{T} = \{X \in L^1(\Theta) : X \text{ is nondecreasing}, X \prec X_{\text{PAM}}\},$$

where X_{PAM} is the interim allocation rule corresponding to a *positive assortative* assignment, formally defined as follows. Let $F^{-1}(q)$ denote the lowest object type $\omega \in \Omega$ such that the total arrival rate of all objects with type weakly lower than ω is at least q. Then, define

$$X_{\text{PAM}}(\theta) = \begin{cases} F^{-1}(G(\theta)) & \text{for } \theta \ge G^{-1}(1-N), \\ \omega_0 & \text{for } \theta < G^{-1}(1-N). \end{cases}$$

[15] By $L^1(\Theta)$ we denote the set of all nonnegative Lebesgue integrable functions with support Θ.
[16] By $u_1(\theta, x)$ we denote the partial derivative of u with respect to θ.

To characterize the optimal solution to (4.1), we note that the functional that we maximize there is convex in X, due to Assumption 2.2 (convexity of u_1 in its second argument). Applying Bauer's Maximum Principle [Bauer, 1958], we then imply that the functional attains its maximum at an extreme point of $\mathcal{T}$. Finally, we use the following extreme point characterization result of Kleiner et al. [2020], translated to our setup, to conclude that the interim allocation X^* that maximizes the above optimization problem has an *interval structure*, which we show corresponds to a monotone disjoint queue mechanism.

Theorem 1 of Kleiner et al. [2020]. *$X : \Theta \to [\omega_0, \omega_n]$ is an extreme point of $\mathcal{T}$ if and only if there exists a collection of disjoint intervals $[\underline{\theta}_i, \bar{\theta}_i)$ indexed by $i \in I$ such that for a.e. $\theta \in \Theta$,*

$$X(\theta) = \begin{cases} X_{\text{PAM}}(\theta), & \theta \notin \cup_{i \in I} [\underline{\theta}_i, \bar{\theta}_i) \\ \frac{\int_{\underline{\theta}_i}^{\bar{\theta}_i} X_{\text{PAM}}(s)\, \mathrm{d}G(s)}{G(\bar{\theta}_i) - G(\underline{\theta}_i)}, & \theta \in [\underline{\theta}_i, \bar{\theta}_i). \end{cases}$$

When the utility function is separable, i.e., $u(\theta, \omega) = \theta\omega$, in Appendix E we describe a constructive approach for finding the optimal mechanism, which is based on ironing techniques, and illustrate it through Example E.1. However, this constructive approach is not applicable when the utility function is not separable. For the broader class of supermodular utilities that we consider in this paper, we next briefly discuss the complications that arise in alternative solution approaches involving classic ironing methods.

Remark 4.3 (Why classic ironing approaches are not directly applicable)**.** The virtual payoff function $\phi(\theta, x)$ in our problem depends on both the agent type θ and the expected object type x through the function $u_1(\theta, x)$. This is in contrast to standard settings (e.g., see [Hartline and Roughgarden, 2008, Condorelli, 2012] for social welfare and [Myerson, 1981] for revenue maximization) where the virtual payoff function depends only on θ. Toikka [2011] developed a method of two-dimensional ironing in the case of continuously differentiable functionals $J(\theta, x)$ that are weakly concave in the allocation x, by constructing a distinct ironed function for each fixed x; i.e., ironing is applied allocation-by-allocation. However, his unconstrained, control-free formulation does not allow for any form of capacity constraints. Otherwise, his pointwise maximization argument would give rise to solutions that do not necessarily satisfy the capacity constraints and thus are infeasible. Adopting his method in our setting would essentially require infinite capacities $F(\omega)$ for every object type $\omega \in \Omega$. Therefore, it does not apply in our problem because of the finite capacities.

4.2 The case of separable utilities

When the utility function is separable, the planner's problem (4.1) turns into maximizing a linear functional. Exploiting this feature, we investigate the structure of monotone disjoint queue mechanisms and discuss their construction for the case of separable utilities.

We say that a monotone disjoint queue mechanism $\mathcal{M}$ *pools only high object types* if it involves $k+1$ queues $q_0, \ldots, q_k$ such that all objects of type ω_i are (i) sent to q_i if $i \leq k-2$, (ii) sent to q_k if $i \geq k$, and (iii) sent to q_{k-1} or q_k if $i = k-1$. Intuitively, such a mechanism *pools* a subset of the highest object types into one queue, and sends every lower object type to a distinct queue.

Similarly, we say that a monotone disjoint queue mechanism $\mathcal{M}$ *pools only low object types* if it involves $k+1$ queues $q_0, \ldots, q_k$ such that all objects of type ω_i are (i) sent to q_0 or q_1 if $i = 0$, (ii) sent to q_1 if $1 \leq i \leq n-k$, (iii) sent to q_1 or q_2 if $i = n-k+1$, and (iv) sent to q_{i-n+k} if $i \geq n-k+2$. Intuitively, such a mechanism *pools* a subset of the lowest object types into one queue, and sends every higher object type to a distinct queue.

A function $z : [0, \bar{\theta}] \to \mathbb{R}$ is *single-peaked* (*single-dipped*) if there is $\theta^* \in \Theta$ such that z is increasing (decreasing) over $[0, \theta^*]$ and decreasing (increasing) over $[\theta^*, \bar{\theta})$. Let $h_G(\theta) = \frac{g(\theta)}{1-G(\theta)}$ denote the hazard rate of G.

Proposition 4.4. *If h_G is single-peaked (single-dipped), then there is an optimal monotone disjoint queue mechanism that pools only low (high) object types.*

We remark that h_G is single-peaked for commonly known distributions G such as Beta, Cauchy, Gamma, Log-normal, Normal, and Weibull. The same holds when these distributions are truncated to have a nonnegative support. Optimal monotone disjoint queue mechanisms thus pool only low object types when G is such a distribution.

The proof of the above proposition uses a constructive ironing technique and is presented in Appendix E (where we also discuss the construction of the optimal mechanism generally in the separable case). To see the intuition, observe that the planner's problem (4.1) equals

$$\sup_{X \in \mathcal{T}} \int_0^{\bar{\theta}} X(\theta)/h_G(\theta) \, \mathrm{d}G(\theta)$$

in the separable case. Consider h_G being single-peaked. Then, there exists $\tilde{\theta}$ such that $\frac{1}{h_G(\theta)}$ is decreasing to the left of $\tilde{\theta}$ but increasing to its right. Since $\mathcal{T}$ is the set of all allocation rules that are monotone and majorized by X_{PAM}, then monotonicity binds to the left of $\tilde{\theta}$ and X^* would be constant there. Notably, due to the involved majorization constraint, X^* generally ends up being constant to the left of some $\tilde{\tilde{\theta}}$ where $\tilde{\tilde{\theta}} > \tilde{\theta}$. Appendix E discusses the details.

5 FCFS: Optimality through information design

In this section we show that, when the planner has control over the information revealed to the agents about the objects, the optimal mechanism, $\mathcal{M}^*$, can be implemented in a simple way: by a FCFS wait-list with deferrals for the agents and an information disclosure policy for objects that *pools adjacent object types*.

Before formally presenting this result, we present an overview. In a FCFS wait-list, every agent enters the same queue and waits for an object to be offered to her. If the agent accepts an offer, she departs the market immediately; otherwise, she would wait for the next offer. Every object is offered to agents in the order of their arrival times and is assigned to the agent with the earliest arrival time who accepts the offer.

In a FCFS wait-list, the planner can also decide how much and what kind of information to disclose to agents about an offered object. One possible disclosure policy, e.g., is providing to all agents not the exact object type, but rather an interval $[\omega_l, \omega_r]$ that contains the object type. An information disclosure policy that pools adjacent objects uses a number of such intervals (to disclose information about the objects) which partition $[\omega_0, \omega_n]$.

The main finding of this section is that the optimal mechanism $\mathcal{M}^*$ can be implemented by a FCFS wait-list paired with an information disclosure policy that pools adjacent object types. The interim payoff of every agent type under this implementation equals the interim payoff of the same agent type in $\mathcal{M}^*$. Proving this result requires analyzing a game played between the agents: in the FCFS wait-list, an agent's decision about whether to accept an offered object depends on how those ahead of her make decisions. The proof shows that, in the unique equilibrium, every agent type attains the same payoff as in $\mathcal{M}^*$.

We next formally define the notion of steady-state equilibrium in a FCFS wait-list. After that, we present the main finding in Section 5.2.

5.1 The FCFS policy at the steady state

We refer to an agent with waiting time t as an agent at *position* t (resembling positions in a queue). We use T to denote the set of all positions, $\mathbb{R}_+$.

We limit our definition of the steady-state equilibria of the FCFS policy to those in which agents make decisions deterministically. Therefore, each agent a has a *decision rule* $\mathcal{D}_a(t) \subseteq \Omega$ that determines which object types she would accept when she is at position t. Since the payoff of an agent is defined by her utility minus her waiting cost, and since the agents are maximizing their payoff, then $\mathcal{D}_a(t)$ should be upward-closed, i.e., if it contains an object type, it also contains all higher object types. We use $\mathcal{C}_a(t)$ to denote the lowest object type that the agent accepts at position t, and call $\mathcal{C}_a(t)$ the agent's *cutoff* at t.

Since we focus on steady-state equilibria, the agents' decision rules can naturally depend on the state of the queue as well. We restrict $\mathcal{C}_a : \mathbb{R}_+ \to \mathbb{R}_+$ to being left-continuous everywhere and continuous almost everywhere. Left-continuity allows the agent to change her cutoff in a discontinuous way, and continuity almost everywhere allows for infinitely many such discontinuities in $\mathcal{C}_a$.

As we will see in the definition of the steady-state equilibrium, every agent chooses her decision rule in order to maximize her expected payoff. We limit our definition to steady-state equilibria

in which agents of the same type choose the same payoff-maximizing decision rule. Hence, in the following definition of the steady-state equilibrium, we use $\mathcal{D}_\theta$ to denote the decision rule of the agents of type θ, and $\mathcal{C}_\theta$ to denote their cutoffs.

Definition. A steady-state equilibrium of the FCFS policy is defined by a tuple $(A, b, m, \bar{t}, \{\mathcal{C}_\theta\}_{\theta\in\Theta})$ that satisfies the following properties:

(i) **The densities of agents at all positions are well-defined:** For each $t \in T$, $A(t)$ is a Lebesgue measure defined over Θ with a corresponding density function $a(t, \cdot) : \Theta \to \mathbb{R}_+$. The function $a(t, \theta)$ gives the measure density of agents of type θ at position t. Furthermore, for each $\theta \in \Theta$, $a(\cdot, \theta) : T \to \mathbb{R}_+$ is regular and left-continuous, and its support is in $[0, \bar{t}]$, where $\bar{t} \in \mathbb{R}_+$. The finiteness of $\bar{t}$ ensures that agents wait in queue for a bounded amount of time.[17]

(ii) **The densities of objects offered to agents at every position is well-defined:** $b : T \times \Omega \to \mathbb{R}_+$, and for each $w \in \Omega$, $b(\cdot, \omega)$ is regular, right-continuous, and increasing; furthermore, $b(\bar{t}, \omega) = F(\omega)$. The function $b(t, \omega)$ denotes the measure of objects of type ω that are offered to agents at position t. The fact that $b(\cdot, \omega)$ is increasing means there are fewer objects left to offer to agents with shorter waiting times.[18]

(iii) **The rate of objects matched to every agent type at every position is well-defined:** $m : T \times \Theta \times \Omega \to \mathbb{R}_+$, and for any $t \in T$ and $\omega \in \Omega$, the Lebesgue integral $M(t, \omega) = \int_{\theta\in\Theta} m(t, \theta, \omega)\, \mathrm{d}G(\theta)$ exists. The function $m(t, \theta, \omega)$ denotes the rate of objects of type ω that are assigned to agents of type θ at position t. The existence of the integral ensures that the rate of objects of a type ω that are assigned to agents at a position t is well-defined.

(iv) $\mathcal{C}_\theta : T \to \mathbb{R}_+$ is regular and left-continuous for all θ.

(v) The *balance equations*, the *market clearing* condition, and *optimality of agents' decisions* hold, which are defined as follows.

Balance equations. The balance equation for agents states that for any $t \in T$ and $\theta \in \Theta$, $a(t, \theta) - \sum_{\omega\in\Omega} m(t, \theta, \omega) = \lim_{t'\to t^+} a(t', \theta)$. The interpretation is that agents do not depart the market unless they are allocated an object. The balance equation for objects states that for any $t \in T$ and $\omega \in \Omega$, $b(t, \omega) - M(t, \omega) = \lim_{t'\to t^-} b(t', \omega)$. The interpretation is that objects remain available until they are allocated to an agent.

[17] The assumption that such $\bar{t}$ exists is without loss of generality, since agents with a negative expected payoff would not join the queue.

[18] We remark that there will never be a shortage of objects of type ω_0: agents can leave the queue with an object of type ω_0 at any time they decide to. This holds because the arrival rate of objects of type ω_0 equals the arrival rate of agents, 1.

Market clearing. We first introduce some notation. Let $\tilde{A}(t,\omega) = \{\theta : \theta \in \Theta, \mathcal{C}_\theta(t) \leq \omega\}$. $\tilde{A}(t,\omega)$ denotes the set of agent types with waiting time t who accept only objects of type at least ω. We suppose that $\tilde{A}(t,\omega)$ is Lebesgue-measurable for all ω, and denote its $A(t)$-measure by $D(t,\omega)$; this is the *aggregate demand* of agents with waiting time t for objects of type at least ω. Since Ω is discrete, then the set $\{\theta : \theta \in \Theta, \mathcal{C}_\theta(t) = \omega\}$ is Lebesgue-measurable as well; we denote its Lebesgue measure by $d(t,\omega)$.

We define the *market clearing* condition by imposing the following conditions on the function $m(t,\theta,\omega)$:

(i) *m respects the supply of objects:* for any $t \in T$ and $\omega \in \Omega$, $\int_{\theta\in\Theta} m(t,\theta,\omega)\,\mathrm{d}\theta$ exists and is at most $b(t,\omega)$;

(ii) *m respects the density of agents:* for $t \in T$ and $\theta \in \Theta$, $\sum_{w\in\Omega} m(t,\theta,\omega) \leq a(t,\theta)$;

(iii) *m respects the agents' decisions:* $m(t,\theta,\omega) > 0$ only if $\mathcal{C}_\theta(t) \geq \omega$. Furthermore, if $\int_{\theta\in\Theta} m(t,\theta,\omega)\,\mathrm{d}\theta < b(t,\omega)$ for some $\omega \in \Omega$, then there exists no type θ with $\mathcal{C}_\theta(t) \leq \omega$ such that $\sum_{\omega\in\Omega} m(t,\theta,\omega) < a(t,\theta)$; and

(iv) *m is agnostic to agents' types and treats agents symmetrically:* for all $\omega \in \Omega_+$, and for all $\theta, \theta' \in \Theta$ such that $\mathcal{C}_\theta(t) \leq \omega$ and $\mathcal{C}_{\theta'}(t) \leq \omega$, $\frac{m(t,\theta,\omega)}{m(t,\theta',\omega)} = \frac{a(t,\theta)}{a(t,\theta')}$. Also, the probability that an agent of type θ at position t is assigned an object ω is $m(t,\theta,\omega)/a(t,\theta)$.

Remark 5.1. *Proposition F.1 in the appendix shows that, despite the broad range of possibilities for the choice of m, this function turns out to be the trivial one at a steady-state equilibrium: at every position t, there would be at most one object type $\omega^t \in \Omega$ for which there is demand and supply, i.e., $b(t,\omega^t), D(t,\omega^t) > 0$. Furthermore, the demand equals the supply for such ω^t, i.e., $b(t,\omega^t) = d(t,\omega^t) > 0$. Hence, the function m assigns all objects of type ω^t that are available at position t to all agents at position t who accept such objects, and makes no other assignments at that position.*

Optimality of an agent's decision. Recall that the decision rule of an agent is a regular and left-continuous function from T to $\mathbb{R}_+$. Let $\mathfrak{C}$ be the set of all such functions. The payoff of an agent with type θ under a decision rule $\mathcal{C} \in \mathfrak{C}$ is $\pi_\theta(\mathcal{C}) = u(\theta,\omega_\mathcal{C}) - t_\mathcal{C}$, where $\omega_\mathcal{C}, t_\mathcal{C}$ respectively denote the agent's expected object type and expected waiting cost under the decision rule $\mathcal{C}$.[19] At a steady-state equilibrium, every agent of type θ chooses a decision rule $\mathcal{C}_\theta \in \mathfrak{C}$ that maximizes her payoff.

This completes the definition of a steady-state equilibrium. We next prove its uniqueness.

Proposition 5.2. *The FCFS policy has a unique steady-state equilibrium.*

[19] We note that $\omega_\mathcal{C}, t_\mathcal{C}$ are well-defined for any $\mathcal{C} \in \mathfrak{C}$.

The unique equilibrium of the FCFS policy can have a lower social welfare than the optimal mechanism $\mathcal{M}^*$, as demonstrated in Example F.3 in the appendix. In fact, the FCFS policy has an outcome equivalent to any incentive-compatible and individually rational that is *positive assortative*, as we formally show next. After that, in Section 5.2, we will show that social welfare under the FCFS policy can reach the optimal level if the planner can design the information disclosed to agents about the object types.

A mechanism $\mathcal{M}$ is *positive assortative* if under $\mathcal{M}$ there can be no two agents a, a' respectively of types θ, θ' such that $\theta > \theta'$ but the object assigned to a has a higher type than the object assigned to a'. The next proposition shows that mechanisms that implement a positive assortative assignment, including priority-based mechanisms, are equivalent to FCFS (with full information disclosure about object types).

Proposition 5.3. *Let $\mathcal{M}$ be an individually rational and incentive-compatible mechanism. Then, if $\mathcal{M}$ is positive assortative, the payoff of every agent in $\mathcal{M}$ equals her payoff in the unique steady-state equilibrium of FCFS.*

We note that our findings indicate that the allocation under the optimal mechanism may be not positive assortative.

5.2 FCFS: Optimality through information design

An *information disclosure policy* μ consists of a finite signal realization space $\widehat{\Omega}$ and a family of distributions $\{\mu(\cdot|\omega)\}_{\omega\in\Omega}$ over $\widehat{\Omega}$. For any information disclosure policy μ, the planner can run the FCFS policy *under* the information disclosure policy μ as follows: for every arriving object of type ω, a single realization is drawn from $\mu(\cdot|\omega)$ independently, which would be revealed to any agent to whom the object is offered.

We say an information disclosure policy μ is *fully revealing* if $\widehat{\Omega} = \Omega$ and, for every $\omega \in \Omega$, $\mu(\cdot|\omega)$ is the degenerate distribution with support $\{\omega\}$. We say a policy μ *pools adjacent object types* if there exists a sequence $\delta_0, \delta_1, \ldots, \delta_k, \delta_{k+1}$ with $\delta_0 = \omega_0$ and $\delta_{k+1} = \omega_n$ such that:

(i) for every nonnegative integer $i \leq k$, there is a distinct signal realization s_i corresponding to the interval (δ_i, δ_{i+1}). For all object types $\omega \in (\delta_i, \delta_{i+1})$, $\mu(\cdot|\omega)$ is the degenerate distribution with support $\{s_i\}$; and

(ii) for all positive integers $i \leq k$, if $\delta_i \in \Omega$, then the support of $\mu(\cdot|\delta_i)$ is a subset of $\{s_{i-1}, s'_i, s_i\}$, where s'_i is a signal realization that may be realized only when the object type is δ_i. The support of $\mu(\cdot|\delta_i)$ is a subset of $\{s'_i, s_i\}$ when $i = 0$, and a subset of $\{s_{i-1}, s'_i\}$ for $i = k + 1$.

Intuitively, such a disclosure policy uses a set S of intervals to cover $[\omega_0, \omega_n]$. The intervals in S are disjoint except possibly at their endpoints. If an object type is an interior point of an interval

$I \in S$, then the policy sends the signal I. Otherwise, the object type must be the endpoint of an interval. In that case, the object type is the endpoint of at most two intervals I, I'. Then, the policy may send either of the signals I, I' or fully reveal the object type.

Proposition 5.4. *If μ pools adjacent object types, then the FCFS policy under μ has a unique steady-state equilibrium.*

Theorem 5.5. *There exists an information disclosure policy μ^* that pools adjacent object types such that the payoff of every agent in the optimal mechanism, $\mathcal{M}^*$, equals her payoff in the unique steady-state equilibrium of the FCFS policy under μ^*.*

Therefore, designing the information disclosed about the objects enables a simple implementation of the optimal mechanism. Remarkably, the information disclosure policy does not use "misleading" signals: the signal realization disclosed for every object is an interval that contains the actual object type. In fact, it follows from the proof that these are the same set of intervals that determine the allocation of objects to queues in the optimal optimal disjoint queue mechanism. (Proving this result, however, requires analyzing a game played between the agents, as discussed earlier.)

An immediate corollary of the latter fact and Proposition 4.4 is that the hazard rate of the distribution of the agents' types G determines the structure of the optimal information disclosure policy, when the agents' utility function u is separable. Particularly, when the hazard rate of G is single-peaked (single-dipped), then the optimal disclosure policy is *lower censorship* (*upper censorship*); i.e., it pools a subset of the lowest (highest) object types together, and fully reveals the type of every higher (lower) object type.

6 Discrete markets

We next show that our main findings hold also in discrete markets where flow arrivals of agents and objects are replaced with Poisson arrivals. Section 6.1 provides a counterpart for Theorem 4.2 (optimality of monotone disjoint queue mechanisms) in sufficiently large discrete markets, and Section 6.2 provides simulations in small markets. Finally, Section 6.3 discusses Theorem 5.5 (optimality of FCFS through information design) in discrete markets.

We next set up a discretized version of our model. Agents and objects arrive to the market over time, starting at time 0, according to independent Poisson processes with rates λ and λN, respectively. If an object is not assigned upon its arrival to an agent, the object will be discarded immediately.[20] The objects' and agents' types are drawn i.i.d. from distributions F and G, respectively.

[20] We run simulations in Section 6.2 that dismiss this assumption.

Similar to our main setup, the agents' and the objects' arrival rates are common knowledge, as well as the queue lengths. An arriving agent observes the queue lengths and then reports a type to the mechanism. The mechanism, in turn, promises the agent an allocation timeline. As the queue lengths are observable, the agent computes the expected waiting cost she would incur from joining each queue, and reports a type to the mechanism that maximizes her expected payoff.[21] After the agent makes the type report, she remains in the pool until she is assigned an object. The difference with the main setup is that the allocation timeline promised by the mechanism may also depend on the history (i.e., the sequence of all events occurred) prior to the agent's arrival.

Incentive-compatible mechanisms are those in which the expected payoff of every participating agent is maximized when she reports her true type to the mechanism. An incentive-compatible mechanism is called *individually rational* if the expected payoff of every participating agent is nonnegative when she reports her true type to the mechanism.

In this discrete setup we focus on analyzing *steady-state* mechanisms. Loosely speaking, we say a mechanism $\mathcal{M}$ is a steady-state mechanism if the distribution of the expected object type assigned by $\mathcal{M}$ to an agent of type θ converges to a well-defined distribution $Y_{\mathcal{M}}(\theta)$ as time approaches infinity. The formal definition is relegated to the appendix (Definition G.1) and requires the mean of $Y_{\mathcal{M}}(\theta)$ to satisfy an additional integrability condition.

6.1 Optimality of disjoint queue mechanisms in large discrete markets

Monotone disjoint queue mechanisms are defined as follows. There are k queues, namely $1, \ldots, k$. Every object of type $\omega \in \Omega_+$ immediately upon arrival is sent to queue i independently with probability $p_{\omega,i}$, where $\sum_{i=1}^{k} p_{\omega,i} = 1$. This is done so that for any two objects of distinct types, the higher-type object is not sent to a lower queue than the queue to which the lower-type object is sent. (Thus, there is positive sorting in the allocation of objects to queues.) The queue lengths at every time and the probabilities $p_{\omega,i}$ for $\omega \in \Omega_+$ and $i \in \{1, \ldots, k\}$ are common knowledge.

When an object is sent to a queue, it is assigned to the agent with the longest waiting time present in that queue. If there is no agent waiting in that queue, the object is discarded. Every agent, upon arrival, decides which queue to join based on her type and her expected waiting cost at each queue. We suppose that the waiting costs are linear: $c(t) = \gamma t$ for a constant $\gamma > 0$. Consider an arriving agent of type θ and suppose that queue i has length l_i upon her arrival. Also, let $\bar{\omega}_i$ denote the expected object type sent to queue i. If the agent joins queue i, then her expected waiting time until receiving an object equals $\frac{l_i+1}{\beta_i}$, where β_i is the objects' arrival rate to queue i. Thus, the agent's expected payoff from joining queue i equals $u(\theta, \bar{\omega}_i) - \gamma \frac{l_i+1}{\beta_i}$. Let i^* denote the queue from which the agent attains the highest expected payoff. If there are multiple such queues,

[21] As in the main setup, the payoff of an agent is defined, as in the main setup, to be her utility from the object she is allocated minus her waiting cost from waiting in the queue. The social welfare in a mechanism is the average of the participating agents' payoffs.

then let i^* be the one with the highest index. The agent joins queue i^* if her payoff at i^* is positive, and otherwise departs the market immediately.

For the following result, we let λ vary while holding the rest of the parameters (N, F, G, γ) fixed. For a given λ, define the *second-best* social welfare, denoted by $W^*(\lambda)$, as the supremum of the social welfares attainable under all incentive-compatible individually rational mechanisms. Also, let $\tilde{W}(\lambda)$ denote the supremum of the social welfare values attainable under all monotone disjoint queue mechanisms.

Proposition 6.1. $\lim_{\lambda\to\infty} \tilde{W}(\lambda) = \lim_{\lambda\to\infty} W^*(\lambda)$.

That is, the optimal monotone disjoint queue mechanism attains the second-best social welfare as the market grows large. In fact, the proof shows that the optimal mechanism $\mathcal{M}^*$ that we find for the continuum model in Section 4 attains the second-best social welfare in the large market limit here.

More precisely, the proof transforms $\mathcal{M}^*$ to a mechanism for the discrete setting as follows: if $\mathcal{M}^*$ sends objects of type ω to queue i with rate $r_{\omega,i}$, then in the discrete setting set $p_{w\,i} = \frac{r_{\omega,i}}{\sum_j r_{\omega,j}}$. We show that the resulting mechanism (in the discrete setting) attains the second-best in the large market limit. The main step of the proof is applying a result from Ashlagi et al. [2020], who provide concentration bounds for the queue lengths in a discrete market.

We next investigate the speed of convergence in the above proposition through simulations. Our simulations also demonstrate that an analogue of Theorem 5.5 (which establishes the optimality of FCFS through information design) holds in discrete markets.

6.2 Simulations for relatively small markets

We next run simulations to demonstrate that the above limit results of this section are meaningful even for small λ. We consider a disjoint queue mechanism with two queues in the discretized setup of Section 6.1 and, using simulations, evaluate the speed of the convergence of its steady state to that of the continuum model, as λ grows large. These simulations indicate that, even for relatively small values of λ, the social welfare under the discretized model sharply approximates the social welfare under the continuum model.

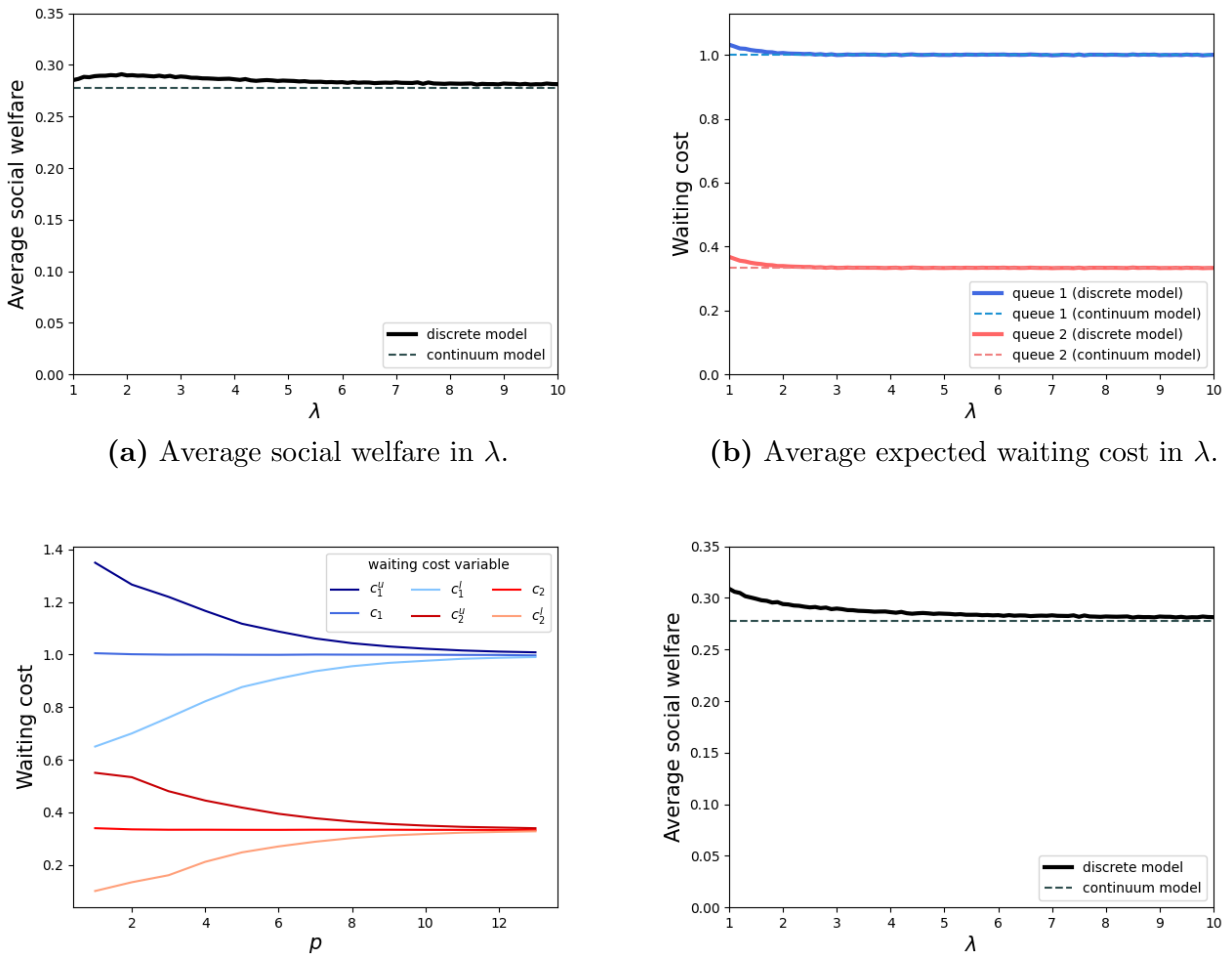
Recall that the type of every agent is drawn i.i.d. from a distribution G; let G be the uniform distribution over the unit interval. Suppose that objects arriving to queue i have an expected type of $\bar{\omega}_i = i$ and an arrival rate of λ, for all $i \in \{1, 2\}$. Agents arrive according to a Poisson process with rate 3λ. (Thus, the arrival rate of agents, 3λ, is larger than the total arrival rate of the objects, 2λ.)

We set $\gamma = 0.05$ and vary λ from 1 to 10 with increments of 0.1. For every such λ, we run the process until 10^7 agents arrive to the market, and then compute the average payoff of these agents. The result is plotted in Figure 3a. In addition, Figure 3b plots the expected cost that an agent incurs if she joins each of the queues upon her arrival.

We observe that the social welfare and the expected waiting cost at each queue approach their counterpart values in the continuum model where agents arrive with a flow rate of 3λ and objects with expected type $\bar{\omega}_i$ arrive to queue i with a flow rate of λ, for $i \in \{1, 2\}$. In particular, observe that all agents of type belonging to $(1/3, 2/3)$ join queue 1, and all agents of type higher than $2/3$ join queue 2. Thus, due to incentive compatibility, the waiting costs at queues $1, 2$ respectively are $1/3$ and 1 in the continuum model. Hence, the social welfare equals

$$\frac{1}{3}\left(2(\frac{2}{3}+1)/2 - 1 + (\frac{1}{3}+\frac{2}{3})/2 - 1/3\right) = \frac{5}{18} \approx 0.2777,$$

as marked in Figure 3a.



(a) Average social welfare in λ.

(b) Average expected waiting cost in λ.

(c) Concentration bounds: $\lambda = 10 \times 2^p$.

(d) Average social welfare: objects are queued.

Figure 3

Figure 3c shows that the expected waiting costs that an agent faces from joining each queue

upon her arrival are sharply concentrated around their means. In other words, at any time the length of each queue is sharply concentrated around its average length. This is demonstrated in the figure by plotting, for every arrival rate λ, an interval $[c_i^l, c_i^u]$ for each queue i. The interval is chosen so that, for 95% of the agents, it contains the waiting cost that the agent expects to incur if she joins queue i (i.e., the interval excludes 2.5% of the data points from each tail of the distribution). We observe that as λ increases, the length of the interval shrinks. This suggests that the waiting cost that an agent expects to incur if she joins queue i is concentrated around its mean c_i, as demonstrated in Figure 3c.

Finally, we consider a slightly different discrete setting where, if an object arrives to a queue when that queue is empty, the object is not discarded. Instead, we assume that such objects are preserved for the agents who choose to join that queue in the future, and will be assigned to these agents in a first-come, first-served manner. Thus, agents assigned to such objects have a waiting time of zero. We make similar observations in this setting. In particular, in Figure 3d we observe that the social welfare approaches the social welfare under the continuum model as λ grows large (an observation analogous to Figure 3a).

6.3 FCFS and information design in discrete markets

We next demonstrate the optimality of FCFS through information design in large discrete markets (an analogue for Theorem 5.5). We prove that the same information disclosure policy μ^* from the continuum setting can attain the optimal level of social welfare, if a certain concentration bound (akin to the one in Figure 3c) holds. Instead of proving the bound formally, we provide simulations that demonstrate that the concentration bound holds. Thus, our argument uses simulations only to verify the concentration bound. The complete argument involves defining a game played between the agents and its equilibria. Due to the technicalities involved, we present the complete argument in Appendix G.2, and summarize it below.

Recall that the information disclosure policy μ^* of Theorem 5.5 consists of the finite signal realization space $\widehat{\Omega}$ and the family of distributions $\{\mu^*(\cdot|\omega)\}_{\omega\in\Omega}$ over $\widehat{\Omega}$. We will use the same disclosure policy but in the discrete setup defined in this section. Define a game between agents by allowing every agent to take an *action* at any time. An action is a threshold belonging to $\widehat{\Omega}\cup\{\infty\}$, which determines the lowest signal realization that the agent accepts if offered at the time of taking the action.

Similar to our main setup, the agents' and the objects' arrival rates and the queue lengths are common knowledge. In addition, an agent can observe the actions of others ahead of her in the FCFS queue. In the equilibrium notion that we define we allow an agent to choose different actions across time. At the particular equilibrium that we construct, however, this does not happen: every agent upon her arrival chooses an action that maximizes her expected payoff and sticks to that action until she is assigned an object, assuming that all other agents behave similarly. We show

that, if the concentration bound (G.1) holds, then the described behavior induces an ϵ-*equilibrium* for arbitrarily small $\epsilon > 0$, when λ is sufficiently large. By an ϵ-equilibrium we mean that the fraction of agents who benefit by deviating from their equilibrium strategy is at most ϵ.

The idea is that, by choosing an action, an agent effectively enters a queue together with all the other agents who have chosen the same action. The way objects are assigned to agents in these queues is slightly different than in monotone disjoint queue mechanisms. Yet, we conjecture that the queue lengths remain sharply concentrated around their lengths in the continuum setting, in the sense of (G.1). Our simulations in the appendix confirm this (Figure 7).

Assuming that (G.1) holds, Proposition G.6 shows that in the discrete model an agent's incentive to join the same queue as in the continuum model is *perturbed* only if the agent's type is arbitrarily close to an *indifferent* agent type as λ grows large. An indifferent agent type is a type θ in the continuum model such that agents of type θ are indifferent over joining at least two of the queues in the optimal monotone disjoint queue mechanism. As the set of indifferent agent types has measure zero, the described behavior results in an ϵ-equilibrium.

7 Conclusion

This paper considers the assignment of objects with different qualities that arrive over time to agents with private information and common waiting costs. The optimal mechanism offers a menu of FCFS queues without deferrals, each of which offers a lottery over adjacent types of objects.

Even simpler, together with disclosing partial information, the optimal mechanism can be implemented using a single FCFS wait-list with deferrals. Such FCFS or priority-based wait-list mechanisms are ubiquitous, for example in the allocation of cadaver organs.

A couple of our assumptions are important. First, agents have identical commonly known cost functions. Second, agents' preferences are vertical and agree on quality. It is intriguing to study settings in which preferences exhibit both horizontal idiosyncrasies (as in Arnosti and Shi [2020]) and a vertical component.

Several concerns may arise in practice. Although FCFS seems simple, it may be difficult for agents to reason about the beliefs of other agents ahead of them in the wait-list. However, a social planner can disclose average waiting times for different (possibly pooled) types of objects. Another concern is the ability to conceal information from agents. This gives rise to the study of optimal mechanisms given constraints over information disclosure policies.

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A Preliminaries

For a positive integer i, denote by $[i]$ the set $\{1, \ldots, i\}$. Let F_0 be the distribution with support Ω_+ that assigns a probability mass $F_0(\omega) = \frac{F(\omega)}{N}$ to every ω in its support.

Consider a mechanism $\mathcal{M}$. The *reduced-form representation* of $\mathcal{M}$ is defined as follows. For an agent of type θ, let $X_{\mathcal{M}}(\theta)$ denote the interim expected object type assigned to her by the mechanism. The function $X_{\mathcal{M}}$ is called the *interim allocation rule*. Also, let $t_{\mathcal{M}}(\theta)$ be the interim expected waiting cost incurred by the agent in $\mathcal{M}$. The function $t_{\mathcal{M}}$ is called the *interim payment rule*. The tuple $(X_{\mathcal{M}}, t_{\mathcal{M}})$ is the *reduced-form representation of* $\mathcal{M}$.

Define $v_{\mathcal{M}}(\theta, \theta') = u(\theta, X_{\mathcal{M}}(\theta'))$, i.e., $v_{\mathcal{M}}(\theta, \theta')$ is the interim expected utility of an agent θ who reports θ' to the mechanism. Similarly, define

$$\Phi_{\mathcal{M}}(\theta, \theta') = v_{\mathcal{M}}(\theta, \theta') - t_{\mathcal{M}}(\theta'),$$

which is the interim expected payoff of an agent of type θ who reports θ' to the mechanism. Let $V_{\mathcal{M}}(\theta) = \Phi_{\mathcal{M}}(\theta, \theta)$ be the interim expected payoff of an agent of type θ who reports her type truthfully.

We note that the reduced-form representation of $\mathcal{M}$ satisfies the interim incentive compatibility and individual rationality constraints, i.e.,

$$\Phi_{\mathcal{M}}(\theta, \theta) \geq \Phi_{\mathcal{M}}(\theta, \theta'), \forall \theta, \theta' \in \Theta \tag{A.1}$$

$$\Phi_{\mathcal{M}}(\theta, \theta) \geq 0, \forall \theta \in \Theta. \tag{A.2}$$

With a slight abuse of notation, we may sometimes drop the subscript $\mathcal{M}$, when it is clear from the context.

B Equivalence of mechanisms

Here we establish standard results about the equivalence of mechanisms that have the same allocations.

Definition B.1. *For two incentive-compatible mechanisms $\mathcal{M}, \mathcal{M}'$ we say $X_{\mathcal{M}}, X_{\mathcal{M}'}$ are equal almost everywhere if $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}'}(\theta)$ holds for all but a finite number of $\theta \in \Theta$.*

Definition B.2. *We say two mechanisms $\mathcal{M}, \mathcal{M}'$ are equivalent if $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}'}(\theta)$ holds almost everywhere and $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}'}(\theta)$ holds for every $\theta \in \Theta$.*

Definition B.3. *We say that a mechanism $\mathcal{M}$ is* regular *if there exists a positive integer k and $\{\theta_0, \ldots, \theta_k\}$, where $0 = \theta_0 \leq \theta_1 \leq \ldots \leq \theta_k = \overline{\theta}$, such that $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}}(\theta')$ holds for any $\theta, \theta' \in (\theta_i, \theta_{i+1})$ where $i < k$ is a nonnegative integer.*

Theorem B.4. *Two regular mechanisms $\mathcal{M}, \mathcal{M}'$ are equivalent if $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}'}(\theta)$ holds everywhere but possibly a measure zero set of $\theta \in \Theta$ and $V_{\mathcal{M}}(0) = V_{\mathcal{M}'}(0)$ for all $\theta \in \Theta$.*

Proof. To prove the claim, we show that $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}'}(\theta')$ for all θ. By the regularity of $\mathcal{M}, \mathcal{M}'$, and by the fact that $X_{\mathcal{M}}, X_{\mathcal{M}'}$ are equal almost everywhere, there exists $\{\theta_0, \ldots, \theta_k\}$, where $0 = \theta_0 \leq \theta_1 \leq \ldots \leq \theta_k = \overline{\theta}$, such that $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}'}(\theta')$ holds for any $\theta, \theta' \in (\theta_i, \theta_{i+1})$ where $i < n$ is a nonnegative integer. Hence, $X_{\mathcal{M}'}(\theta) = X_{\mathcal{M}'}(\theta')$ also holds for any $\theta, \theta' \in (\theta_i, \theta_{i+1})$ where $i < n$ is a nonnegative integer.

On the other hand, we must have $t_{\mathcal{M}}(\theta) = t_{\mathcal{M}}(\theta')$ for any $\theta, \theta' < \theta_1$. Therefore, $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}}(\theta')$ holds for any $\theta, \theta' < \theta_1$. Similarly, $V_{\mathcal{M}'}(\theta) = V_{\mathcal{M}}(\theta')$ holds for any $\theta, \theta' < \theta_1$. Therefore, we have $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}'}(\theta)$ for any $\theta < \theta_1$.

Next, we will show that $V_{\mathcal{M}}(\theta_1) = V_{\mathcal{M}'}(\theta_1)$. Let $\theta_l \in (\theta_0, \theta_1)$ and $\theta_r \in (\theta_1, \theta_2)$. Since $\mathcal{M}$ is incentive-compatible,

$$\begin{cases} V_{\mathcal{M}}(\theta_l) = \theta_l X_{\mathcal{M}}(\theta_l) - t_{\mathcal{M}}(\theta_l) \geq \theta_l X_{\mathcal{M}}(\theta_r) - t_{\mathcal{M}}(\theta_l) \\ V_{\mathcal{M}}(\theta_r) = \theta_r X_{\mathcal{M}}(\theta_r) - t_{\mathcal{M}}(\theta_r) \geq \theta_r X_{\mathcal{M}}(\theta_l) - t_{\mathcal{M}}(\theta_r). \end{cases}$$

The above inequalities imply that

$$\lim_{\theta_l \to \theta_1^-} \theta_l X_{\mathcal{M}}(\theta_l) - t_{\mathcal{M}}(\theta_l) = \lim_{\theta_r \to \theta_1^+} \theta_r X_{\mathcal{M}}(\theta_r) - t_{\mathcal{M}}(\theta_r) = \theta_1 X_{\mathcal{M}}(\theta_1) - t_{\mathcal{M}}(\theta_1). \tag{B.1}$$

Since we have shown that

$$V_{\mathcal{M}}(\theta_l) = \theta_l X_{\mathcal{M}}(\theta_l) - t_{\mathcal{M}}(\theta_l) = \theta_l X_{\mathcal{M}'}(\theta_l) - t_{\mathcal{M}'}(\theta_l) = V_{\mathcal{M}'}(\theta_l),$$

then (B.1) implies that $V_{\mathcal{M}}(\theta_1) = V_{\mathcal{M}'}(\theta_1)$. Thus, we have proved the claim of the theorem for all $\theta \leq \theta_1$. The proof for $\theta > \theta_1$ follows inductively: given $V_{\mathcal{M}}(\theta_1) = V_{\mathcal{M}'}(\theta_1)$, we show that $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}'}(\theta)$ holds for all $\theta \in (\theta_1, \theta_2)$, as we did in the first step of the proof for $\theta \in (\theta_0, \theta_1)$. Then, we use this fact to prove $V_{\mathcal{M}}(\theta_2) = V_{\mathcal{M}'}(\theta_2)$, as we proved this equality for θ_1 in the second step of the proof. Continuing this argument proves that $V_{\mathcal{M}}(\theta) = V_{\mathcal{M}'}(\theta)$ for all $\theta \leq \theta_k$. □

C Proofs for Section 4

Proof of Proposition 4.1. Let the monotone disjoint queue mechanism be defined by the sequence $\theta_0 = 0, \theta_1, \ldots, \theta_k, \theta_{k+1} = \overline{\theta}$, as we defined in Section 4. The proof idea is that the steady-state queue lengths should be such that an agent of type θ_i is indifferent between joining q_i and q_{i-1}. This will determine the queue lengths uniquely, as formalized below.

Let l_i denote the steady-state length of q_i. Also, let $\overline{\omega}_i$ denote the expected object type that is

assigned to q_i, i.e.,

$$\overline{\omega}_i = \mathbb{E}_{\omega\sim F}\left[\omega \middle| \text{object type } \omega \text{ is allocated to } q_i\right].$$

The payoff of an agent of type θ from joining a queue q_i is then defined by $\pi_i(\theta) = u(\theta, \overline{\omega}_i) - c(l_i)$. By the incentive compatibility constraint and the continuity of $\pi_1(\cdot)$, agents of type θ_1 should be indifferent between joining queue q_1 and being assigned to the object ω_0. Thus, $u(\theta_1, \omega_0) = u(\theta_1, \omega_1) - l_1$ should hold, which means that

$$l_1 = c^{-1}(u(\theta_1, \omega_1) - u(\theta_1, \omega_0)). \tag{C.1}$$

Similarly, the continuity of $\pi_2(\cdot)$ implies that agents of type θ_2 should be indifferent between joining queues q_1 and q_2. Therefore, $\pi_1(\theta_2) = \pi_2(\theta_2)$. This means that

$$u(\overline{\omega}_2, \theta_2) - c(l_2) = u(\overline{\omega}_1, \theta_2) - c(l_1),$$

which implies that

$$l_2 = c^{-1}(u(\overline{\omega}_2, \theta_2) - u(\overline{\omega}_1, \theta_2) + c(l_1)). \tag{C.2}$$

The same argument applies to all $i \geq 2$, which proves the claim. □

D Proof of Theorem 4.2

To characterize the optimal mechanisms, we use majorization theory and properties of convex functionals to reformulate and solve the optimization problem, and show that the optimal solution corresponds to a (non-wasteful) disjoint queue mechanism.

Characterization of the optimal mechanism. We first define the set of *feasible interim allocation rules*. Formally, let $\overline{\Omega} = [\omega_0, \omega_n]$ and

$$\mathcal{S} = \{X : \Theta \to \overline{\Omega} \mid X \text{ is nondecreasing, and } \exists x \in \mathcal{D} \text{ s.t. } X(\theta) = \sum_{\omega\in\Omega} \omega x(\theta, \omega)\} \tag{D.1}$$

where

$$\mathcal{D} = \{x : \Theta \times \Omega \to [0, 1] \mid x(\theta, \cdot) \in \Delta\Omega, \int_0^{\bar{\theta}} x(\theta, \omega)\, \mathrm{d}G(\theta) \leq F(\omega), \forall \omega \in \Omega_+\}. \tag{D.2}$$

The results of Rochet [1987] show that the set of feasible interim allocation rules are also *implementable*, in the following sense. Under Assumption 2.3 and Assumption 2.1, Rochet [1987] shows that a necessary and sufficient condition for $(X_{\mathcal{M}}(\theta), t_{\mathcal{M}}(\theta))$ being truthfully implementable in dominant strategies is $X_{\mathcal{M}}(\theta)$ being nondecreasing in θ.

Lemma D.1 (Proposition 1 in Rochet [1987]). *Let $u \in C^2$ and assume that u has the Spence-Mirrlees property. Then, the interim allocation rule $X_{\mathcal{M}}$ of any incentive-compatible mechanism must be nondecreasing in θ.*

The converse statement also follows from [Rochet, 1987].

Lemma D.2. *If $X_{\mathcal{M}}(\theta)$ is nondecreasing in θ then there exists a nonnegative interim payment rule $t_{\mathcal{M}} : \Theta \to \mathbb{R}_+$ such that $(X_{\mathcal{M}}(\theta), t_{\mathcal{M}}(\theta))$ is induced by an incentive-compatible mechanism $\mathcal{M}$.*

Next, we define the function $\phi(\theta, x)$ and reduce the social welfare optimization problem to the simplified problem of maximizing the expected value of $\phi(\theta, X(\theta))$. The virtual payoff function $\phi : \Theta \times [\omega_0, \omega_n] \to \mathbb{R}$, is defined as

$$\phi(\theta, x) = \frac{1 - G(\theta)}{g(\theta)} \cdot u_1(\theta, x),$$

where $u_1(\theta, x)$ is the partial derivative of u with respect to θ.

Lemma D.3. *The social welfare $W_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies*

$$W_{\mathcal{M}^*} = \sup_{X \in \mathcal{S}} \int_0^{\bar{\theta}} \phi(\theta, X(\theta)) \, \mathrm{d}G(\theta).$$

Proof. By Toikka [2011, Lemma D.4] and Assumption 2.4, the interim payment rule $t_{\mathcal{M}}$ is given by

$$t_{\mathcal{M}}(\theta) = u(\theta, X_{\mathcal{M}}(\theta)) - \int_0^{\theta} u_1(\tau, X_{\mathcal{M}}(\tau)) \, \mathrm{d}\tau, \theta \in \Theta. \tag{D.3}$$

Therefore, the social welfare under $\mathcal{M}$ equals

$$\int_0^{\bar{\theta}} [u(\theta, X_{\mathcal{M}}(\theta)) - t_{\mathcal{M}}(\theta))] \, \mathrm{d}G(\theta) = \int_0^{\bar{\theta}} \int_0^{\theta} u_1(\tau, X_{\mathcal{M}}(\tau)) d\tau \, \mathrm{d}G(\theta).$$

Thus, for the optimal social welfare $W_{\mathcal{M}^*}$, the previous equation gives us—after reversing the integration order due to Fubini's theorem—that

$$W_{\mathcal{M}^*} = \sup_{X \in \mathcal{S}} \int_0^{\bar{\theta}} \frac{1 - G(\theta)}{g(\theta)} u_1(\theta, X(\theta)) \, \mathrm{d}G(\theta) = \sup_{X \in \mathcal{S}} \int_0^{\bar{\theta}} \phi(\theta, X(\theta)) \, \mathrm{d}G(\theta)$$

since we are optimizing over the set of all feasible interim allocation rules $X \in \mathcal{S}$. This proves the lemma. □

Note that Assumption 2.1, together with Assumption 2.4, implies that $u(\theta, x)$ must be increasing in x. Thus, the interim individual rationality constraint is trivially satisfied. (On the other hand, ex-post individual rationality may not be satisfied: some agents may wait for an object but not get

anything in the end.) Furthermore, as it easily follows from Lemma D.1 and the payment formula (D.3), the fact that u is increasing in x guarantees that the expected waiting costs are nonnegative.

Majorization. Due to Lemma D.1, we can focus only on nondecreasing functional maximizers $X(\theta)$. Borrowing the terminology from *majorization* theory (see, e.g., [Kleiner et al., 2020]), for two nondecreasing interim allocation rules $X, Y \in L^1(\Theta)$, we say that X *majorizes* Y, denoted by $Y \prec X$, if

$$\int_{\theta}^{\bar{\theta}} Y(s)\,\mathrm{d}G(s) \leq \int_{\theta}^{\bar{\theta}} X(s)\,\mathrm{d}G(s) \text{ for all } \theta \in \Theta \tag{D.4}$$

$$\int_{0}^{\bar{\theta}} Y(s)\,\mathrm{d}G(s) = \int_{0}^{\bar{\theta}} X(s)\,\mathrm{d}G(s). \tag{D.5}$$

Similarly, for two nondecreasing interim allocation rules $X, Y \in L^1(\Theta)$, we say that X *weakly majorizes* Y, denoted by $Y \prec_w X$, if (D.4) holds (but not necessarily (D.5)).

Furthermore, let

$$F^{-1}(q) = \inf\{\omega \in \Omega : \sum_{\omega' \in \Omega:\ \omega \geq \omega'} F(\omega') \geq q\}, q \in [0, N]$$

denote the lowest object type ω such that the total arrival rate of all objects with type weakly lower than ω is at least q.

We say that an interim allocation rule $X_{\mathcal{M}}$ of an incentive-compatible and individually rational mechanism $\mathcal{M}$ is *non-wasteful* if it is feasible (i.e., $X_{\mathcal{M}} \in \mathcal{S}$) and

$$\int_{0}^{\bar{\theta}} X_{\mathcal{M}}(s)\,\mathrm{d}G(s) = \sum_{\omega \in \Omega_+} \omega F(\omega) + \omega_0(1 - N). \tag{D.6}$$

We say that $X_{\mathcal{M}}$ is *wasteful* if it is feasible but does not satisfy condition (D.6). Later, We show that the optimal mechanism must be non-wasteful (Proposition D.11).

Let X_{PAM} denote the interim allocation rule corresponding to a positive assortative assignment, i.e., $X_{\text{PAM}}(\theta) = F^{-1}(G(\theta))$ for $\theta \geq G^{-1}(1 - N)$ and $X_{\text{PAM}}(\theta) = \omega_0$ for $\theta < G^{-1}(1 - N)$. Observe that X_{PAM} is, by definition, a non-wasteful interim allocation rule.

Lemma D.4. *The set*

$$T_{m,w}(X_{\text{PAM}}) = \{X \in L^1(\Theta) : X \text{ is nondecreasing}, X \prec_w X_{\text{PAM}}\} \tag{D.7}$$

is the set $\mathcal{S}$ of feasible interim allocation rules, both wasteful and non-wasteful, of all incentive-compatible and individually rational mechanisms.

Proof. We first show that if $X \in T_{m,w}(X_{\text{PAM}})$, then X corresponds to the interim allocation rule of an

incentive-compatible and individually rational mechanism. Since $X \in T_{m,w}(X_{\text{PAM}})$ is nondecreasing, then Lemma D.2 applies and determines an agent's waiting time in the allocation timeline assigned to her by the mechanism. It remains to determine the distribution of the object types assigned to the agent by the mechanism, so that the expected object type assigned to her equals $X(\theta)$ when she has type θ. The proof for this part is analogous to Theorem 5 of [Kleiner et al., 2020], by defining a *pushforward* measure that transforms X_{PAM} to X.

Next, we show that if weak majorization does not hold, that is $X \notin T_{m,w}(X_{\text{PAM}})$, then X does not correspond to the interim allocation rule of an incentive-compatible and individually rational mechanism. The proof is by contradiction. Thus, suppose that X does correspond to the interim allocation rule of an incentive-compatible and individually rational mechanism. Let θ^* denote an arbitrary agent type at which the weak majorization property fails. Then,

$$\int_{\theta^*}^{\bar{\theta}} X(s)\,\mathrm{d}G(s) > \int_{\theta^*}^{\bar{\theta}} X_{\text{PAM}}(s)\,\mathrm{d}G(s).$$

Thus, whereas X_{PAM} allocates a mass $1-G(\theta^*)$ of objects of average quality $\overline{\omega} = \int_{\theta^*}^{\bar{\theta}} X_{\text{PAM}}(s)\,\mathrm{d}G(s)$, X allocates an equal mass of objects but with an average quality higher than $\overline{\omega}$. This is a contradiction, since X_{PAM} assigns objects to agents in an assortative manner. The proof is complete. □

Proposition D.5. *The social welfare $W_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies*

$$W_{\mathcal{M}^*} = \sup_{X \in T_{m,w}(X_{\text{PAM}})} \int_0^{\bar{\theta}} \phi(\theta, X(\theta))\,\mathrm{d}G(\theta).$$

Proof. It follows immediately from Lemma D.3 and Lemma D.4. □

Finally, let

$$T_m(X_{\text{PAM}}) = \{X \in L^1(\Theta) : X \text{ is nondecreasing}, X \prec X_{\text{PAM}}\}. \tag{D.8}$$

Note that, by the previous definitions, $T_m(X_{\text{PAM}}) \subseteq T_{m,w}(X_{\text{PAM}})$. As we will finally show in Proposition D.11, the optimal X in Proposition D.5 actually satisfies $X \in T_m(X_{\text{PAM}})$.

Optimization of convex functionals. Next, we use some classic results for maximization of convex functions over convex sets. The key steps are as follows. First, by Proposition 1 in [Kleiner et al., 2020], we know that the sets $T_m(X_{\text{PAM}})$ and $T_{m,w}(X_{\text{PAM}})$ are convex and hence their respective sets of extreme points are non-empty. Second, by Bauer's Maximum Principle [Bauer, 1958], we know that an upper semicontinuous convex real functional h over a compact convex subset K attains its maximum at an extreme point of K. Thus, we can apply this principle to the convex sets $T_m(X_{\text{PAM}})$, $T_{m,w}(X_{\text{PAM}})$ and our social welfare functional assuming that the virtual payoff function $\phi(\theta, x)$ is convex in the allocation argument (by Lemma D.8). The latter condition simply reduces

to $u_1(\theta, \omega)$ being convex in ω. Under this assumption, this proves – together with Proposition D.5 – that virtual welfare is maximized at some extreme point of $T_{m,w}(X_{\text{PAM}})$; in Proposition D.11, we further show that the optimal allocation rule must be non-wasteful, thus that virtual welfare is maximized at some extreme point of $T_m(X_{\text{PAM}}) \subseteq T_{m,w}(X_{\text{PAM}})$. Finally, using Theorem 1 of Kleiner et al. [2020], which guarantees that the extreme points of $T_m(X_{\text{PAM}})$ have the form (D.9), we prove that the optimal $(X_{\mathcal{M}^*}, t_{\mathcal{M}^*})$ satisfies the definition of disjoint queue mechanisms.

Theorem D.6 (Proposition 1 in Kleiner et al. [2020])**.** *The sets $T_m(X_{\text{PAM}})$ and $T_{m,w}(X_{\text{PAM}})$ are convex and compact in the norm topology, and hence the respective sets of extreme points are non-empty.*

Proof. The result follows from Proposition 1 in Kleiner et al. [2020] and the fact that $X_{\text{PAM}} \in L^1(\Theta)$ is nondecreasing. □

Theorem D.7 (Bauer's Maximum Principle, Bauer [1958])**.** *Let $\mathcal{X}$ be a locally convex real space. An upper semicontinuous convex real functional h over a compact convex subset $K \subset \mathcal{X}$ attains its maximum at an extreme point of K,*

$$\sup_{x \in K} h(x) = \max_{\tilde{x} \in \text{Ext}(K)} h(\tilde{x}),$$

where $\text{Ext}(K)$ is the (non–empty) set of extreme points of K.

Lemma D.8. *Assume that the virtual payoff function $\phi(\theta, x)$ where $\phi : \Theta \times \overline{\Omega} \to \mathbb{R}$ is convex in x. Then, the integral functional*

$$\int_0^{\bar{\theta}} \phi(\theta, X(\theta)) \, \mathrm{d}G(\theta)$$

is convex on $T_m(X_{\text{PAM}})$ and $T_{m,w}(X_{\text{PAM}})$.

Proof. The result follows immediately from Theorem 22 (a) in Rockafellar [1974]. □

Theorem D.9 (Theorem 1 in Kleiner et al. [2020])**.** *$X : \Theta \to \overline{\Omega}$ is an extreme point of $T_m(X_{\text{PAM}})$ if and only if there exists a collection of disjoint intervals $[\underline{\theta}_i, \bar{\theta}_i)$ indexed by $i \in I$ such that for a.e. $\theta \in \Theta$,*

$$X(\theta) = \begin{cases} X_{\text{PAM}}(\theta), & \theta \notin \cup_{i \in I} [\underline{\theta}_i, \bar{\theta}_i) \\ \frac{\int_{\underline{\theta}_i}^{\bar{\theta}_i} X_{\text{PAM}}(s) \, \mathrm{d}G(s)}{G(\bar{\theta}_i) - G(\underline{\theta}_i)}, & \theta \in [\underline{\theta}_i, \bar{\theta}_i). \end{cases} \tag{D.9}$$

The following lemma follows immediately from Corollary 2 of Kleiner et al. [2020], and from the fact that X_{PAM} is nondecreasing and nonnegative. To adapt the notation to our setting, we use the definition

$$\chi_{[\theta^*, \bar{\theta})}(X_{\text{PAM}}(\theta)) = \begin{cases} X_{\text{PAM}}(\theta), & \theta \in [\theta^*, \bar{\theta}) \\ \omega_0, & \theta \notin [\theta^*, \bar{\theta}). \end{cases}$$

Lemma D.10 (Corollary 2 in Kleiner et al. [2020]). *$X : \Theta \to \overline{\Omega}$ is an extreme point of $T_{m,w}(X_{\text{PAM}})$ if and only if it is an extreme point of $T_m(\chi_{[\theta^*,\bar{\theta})}(X_{\text{PAM}}))$ for some $\theta^* \in [0, \bar{\theta})$.*

Proposition D.11. *Suppose that $\phi(\theta, x)$ is convex in x. Let $\mathcal{M}^*$ denote the optimal incentive-compatible and individually rational mechanism. Then, its corresponding interim allocation rule $X_{\mathcal{M}^*}$ is non-wasteful, i.e., $X_{\mathcal{M}^*} \in T_m(X_{\text{PAM}})$.*

Proof. By Lemma D.8 and Theorem D.6, we can apply Theorem D.7, which implies that a maximizer $X_{\mathcal{M}^*}(\theta)$ of the expected social welfare $\int_0^{\bar{\theta}} \phi(\theta, X(\theta))\,\mathrm{d}G(\theta)$ over $T_{m,w}(X_{\text{PAM}})$ (respectively, $T_m(X_{\text{PAM}})$) is attained at an extreme point of $T_{m,w}(X_{\text{PAM}})$ (respectively, $T_m(X_{\text{PAM}})$).

By Proposition D.5, the optimal interim allocation rule $X_{\mathcal{M}^*} \in T_{m,w}(X_{\text{PAM}})$; furthermore, $T_m(X_{\text{PAM}}) \subseteq T_{m,w}(X_{\text{PAM}})$. Therefore, it suffices to show that $X_{\mathcal{M}^*}$ cannot be wasteful, i.e., $X_{\mathcal{M}^*} \notin T_{m,w}(X_{\text{PAM}}) \setminus T_m(X_{\text{PAM}})$.

The proof is by contradiction. Suppose that $X_{\mathcal{M}^*}$ is wasteful. By Lemma D.10 and Theorem D.9, $X_{\mathcal{M}^*}$ must have the form

$$X_{\mathcal{M}^*}(\theta) = \begin{cases} \chi_{[\theta^*,\bar{\theta})}(X_{\text{PAM}}(\theta)), & \theta \notin \cup_{i\in I}[\underline{\theta}_i, \bar{\theta}_i) \\ \frac{\int_{\underline{\theta}_i}^{\bar{\theta}_i} \chi_{[\theta^*,\bar{\theta})}(X_{\text{PAM}}(s))\,\mathrm{d}G(s)}{G(\bar{\theta}_i)-G(\underline{\theta}_i)}, & \theta \in [\underline{\theta}_i, \bar{\theta}_i). \end{cases}$$

Since $X_{\mathcal{M}^*}$ is wasteful, it must hold that $G(\theta^*) > 1 - N$. We construct a new interim allocation rule $X_{\mathcal{M}}$ that uses the same collection of disjoint intervals $[\underline{\theta}_i, \bar{\theta}_i)$ but instead of $\chi_{[\theta^*,\bar{\theta})}(X_{\text{PAM}})$ it uses X_{PAM}; i.e.,

$$X_{\mathcal{M}}(\theta) = \begin{cases} X_{\text{PAM}}(\theta), & \theta \notin \cup_{i\in I}[\underline{\theta}_i, \bar{\theta}_i) \\ \frac{\int_{\underline{\theta}_i}^{\bar{\theta}_i} X_{\text{PAM}}(s)\,\mathrm{d}G(s)}{G(\bar{\theta}_i)-G(\underline{\theta}_i)}, & \theta \in [\underline{\theta}_i, \bar{\theta}_i). \end{cases}$$

By construction, $X_{\mathcal{M}} \in T_m(X_{\text{PAM}})$, thus it is feasible and incentive-compatible, and $X_{\mathcal{M}}(\theta) \geq X_{\mathcal{M}^*}(\theta)$ for all $\theta \in \Theta$ and $X_{\mathcal{M}}(\theta) > X_{\mathcal{M}^*}(\theta)$ at least for $\theta \in [G^{-1}(1-N), \theta^*)$, since $X_{\mathcal{M}}(\theta)$ does not waste objects. As a result, social welfare under $\mathcal{M}$ is strictly larger, i.e.,

$$W_{\mathcal{M}^*} = \int_0^{\bar{\theta}} \phi(\theta, X_{\mathcal{M}^*}(\theta))\,\mathrm{d}G(\theta) < \int_0^{\bar{\theta}} \phi(\theta, X_{\mathcal{M}}(\theta))\,\mathrm{d}G(\theta) = W_{\mathcal{M}}.$$

This contradicts the optimality of $X_{\mathcal{M}^*}(\theta)$. Therefore, $X_{\mathcal{M}^*}(\theta)$ must be non-wasteful. □

Theorem D.12. *The optimal mechanism, $\mathcal{M}^*$, is a monotone disjoint queue mechanism.*

Proof. By Proposition D.11, the interim allocation rule $X_{\mathcal{M}^*}$ of $\mathcal{M}^*$ is non-wasteful, that is, $X_{\mathcal{M}^*} \in T_m(X_{\text{PAM}})$.

It suffices to show that any $X_{\mathcal{M}^*} \in T_m(X_{\text{PAM}})$ is associated with a monotone disjoint queue mechanism. Specifically, due to Theorem D.9, there exists an increasing sequence of positive reals $\theta_0 = 0, \theta_1, \ldots, \theta_k, \theta_{k+1} = \bar{\theta}$ such that agents of type $\theta \in [\theta_j, \theta_{j+1})$, $j \geq 0$, get the same allocation

$X_{\mathcal{M}^*}(\theta_j)$. This sequence is finite. Indeed, X_{PAM} is constant everywhere except for $n+1$ points. Thus, (D.9) and the definition of X_{PAM} together imply that there are at most $2n+1$ possible values that $X_{\mathcal{M}^*}$ can take.[22] Consequently, $k \leq 2n-1$.

By (D.3), agents receiving the same allocation $X_{\mathcal{M}^*}(\theta)$ will incur the same waiting cost. Note that for all agents with types $\theta \in [\theta_j, \theta_{j+1})$, $j \geq 0$, it holds that

$$t_{\mathcal{M}^*}(\theta) = \int_0^{\theta} u_2(\theta, X_{\mathcal{M}^*}(\theta)) X'_{\mathcal{M}^*}(\theta)\,\mathrm{d}\theta = \int_0^{\theta_j} u_2(\theta, X_{\mathcal{M}^*}(\theta)) X'_{\mathcal{M}^*}(\theta)\,\mathrm{d}\theta \tag{D.10}$$

since $X'_{\mathcal{M}^*}(\theta) = 0$ for $\theta \in [\theta_j, \theta_{j+1})$. Observe that waiting costs are nonnegative; indeed, by (D.10), we can write $t_{\mathcal{M}^*}(\theta) = 0$ for $\theta \in [\theta_0, \theta_1]$ and for $\theta \in [\theta_j, \theta_{j+1})$, $1 \leq j \leq k$,

$$t_{\mathcal{M}^*}(\theta) = \sum_{i=1}^{j} \left(u(\theta_i, X_{\mathcal{M}^*}(\theta_i)) - u(\theta_i, X_{\mathcal{M}^*}(\theta_{i-1}))\right) > 0;$$

the inequality follows from Assumption 2.1 and Assumption 2.4, which imply that $u(\theta, x)$ is increasing in x.

Finally, we observe that $X_{\mathcal{M}^*}, t_{\mathcal{M}^*}$ are respectively the interim allocation and payment rules for the following monotone disjoint queue mechanism. The mechanism includes $k+1$ queues $q_0, q_1, \ldots, q_k$ where an agent who reports a type $\theta \in [\theta_i, \theta_{i+1})$ for $i \geq 0$ is assigned upon arrival to queue q_i. Furthermore, (D.9) shows that $X_{\mathcal{M}^*}(\theta)$ is piecewise-constant, increasing, and pools only adjacent object types. The steady-state queue lengths in such a mechanism are given by $c^{-1}(t_{\mathcal{M}^*}(\theta_j))$ for q_j. Thus, all conditions for monotone disjoint queue mechanisms are satisfied.

□

D.1 Weighted social welfare

In this subsection, we extend our main results to the more general case where social welfare is defined as a *weighted average* over the agents' payoffs rather than a simple average. Specifically, let $w : \Theta \to \mathbb{R}_+$ be a continuous function assigning a nonnegative weight to each agent type $\theta \in \Theta$. Then, the planner's objective is to maximize the weighted social welfare

$$\int_0^{\bar{\theta}} w(\theta)\left(u(\theta, X_{\mathcal{M}}(\theta)) - t_{\mathcal{M}}(\theta)\right)\mathrm{d}G(\theta)$$

over the set of individually rational and incentive-compatible mechanisms $\mathcal{M}$.

To adapt the proof of Theorem 4.2, we extend the definition of virtual payoff function to

[22] In the *worst* case, $X_{\mathcal{M}^*}$ takes values $\omega_0, \omega_0^1, \omega_1, \ldots, \omega_{n-1}^n, \omega_n$, where ω_i^{i+1} is some value $w_i < \omega_i^{i+1} < \omega_{i+1}$ for $0 \leq i < n$.

$\widehat{\phi} : \Theta \times [\omega_0, \omega_n] \to \mathbb{R}_+$ as follows:

$$\widehat{\phi}(\theta, x) = \mathbb{E}_{\tau \sim G}[w(\tau) \mid \tau > \theta] \frac{1 - G(\theta)}{g(\theta)} u_1(\theta, x)$$

Then, we have the following counterpart to Proposition D.5.

Proposition D.13. *The weighted social welfare $\widehat{W}_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies*

$$\widehat{W}_{\mathcal{M}^*} = \sup_{X \in T_{m,w}(X_{\text{PAM}})} \int_0^{\bar{\theta}} \widehat{\phi}(\theta, X(\theta)) \, \mathrm{d}G(\theta).$$

Proof. Using (D.3), the weighted social welfare objective can be written as

$$\begin{aligned}
\int_0^{\bar{\theta}} w(\theta) \left(u(\theta, X_{\mathcal{M}}(\theta)) - t_{\mathcal{M}}(\theta)\right) \mathrm{d}G(\theta) &= \int_0^{\bar{\theta}} w(\theta) \int_0^{\theta} u_1(\tau, X_{\mathcal{M}}(\tau)) \, \mathrm{d}\tau \, \mathrm{d}G(\theta) \\
&= \int_0^{\bar{\theta}} u_1(\tau, X_{\mathcal{M}}(\tau)) \int_{\tau}^{\bar{\theta}} w(\theta) \, \mathrm{d}G(\theta) \, \mathrm{d}\tau \\
&= \int_0^{\bar{\theta}} \mathbb{E}_{\tau \sim G}[w(\tau) \mid \tau > \theta] \frac{1 - G(\theta)}{g(\theta)} u_1(\theta, X_M(\theta) \, \mathrm{d}G(\theta) \\
&= \int_0^{\bar{\theta}} \widehat{\phi}(\theta, X(\theta)) \, \mathrm{d}G(\theta)
\end{aligned}$$

By Lemma D.4, we finally get that the weighted social welfare $\widehat{W}_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies

$$\widehat{W}_{\mathcal{M}^*} = \sup_{X \in T_{m,w}(X_{\text{PAM}})} \int_0^{\bar{\theta}} \widehat{\phi}(\theta, X(\theta)) \, \mathrm{d}G(\theta).$$

□

Furthermore, since the weight function w is continuous and depends only on agent types $\theta \in \Theta$, and since $\widehat{\phi}(\theta, x)$ is convex in x by Assumption 2.2, the analogous result of Lemma D.8 holds; i.e., the weighted social welfare objective is a functional convex in X. Consequently, Proposition D.11 would still hold when ϕ is replaced with $\widehat{\phi}$ in its statement. Hence, under a weighted social welfare objective, the optimal mechanism, $\mathcal{M}^*$, remains a monotone disjoint queue mechanism.

E Separable utilities: Proof of Proposition 4.4 and the constructive approach

From Proposition D.5 it follows that the social welfare $W_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies

$$W_{\mathcal{M}^*} = \sup_{X \in T_{m,w}(X_{\text{PAM}})} \int_0^{\bar{\theta}} X(\theta)/h_G(\theta) \, \mathrm{d}G(\theta).$$

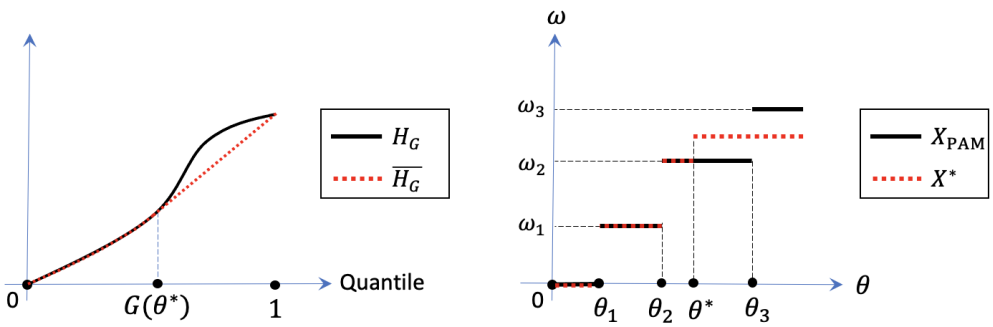

Figure 4: **Left panel:** $\overline{H_G}$ is affine to the right of $G(\theta^*)$ but not to its left. **Right panel:** the optimal rule $X^*(\theta)$ equals $\frac{\int_{\theta^*}^{\bar{\theta}} X_{\text{PAM}}(\kappa)dG(\kappa)}{1-G(\theta^*)}$ at $\theta > \theta^*$ and equals $X_{\text{PAM}}(\theta)$ otherwise. The corresponding monotone disjoint queue mechanism sends objects of type ω_0 to q_0, objects of type ω_1 to q_1 at a rate $F(w_1)$, and objects of type ω_2 to q_2 at a rate $G(\theta^*) - G(\theta_2)$, and pools the remaining fraction of objects of type w_2 with all objects of type ω_3 and sends them together to q_3.

One can solve this problem using classical ironing techniques. In particular, to solve this problem we apply Proposition 2 of [Kleiner et al., 2020], which provides an elegant description of ironing when maximizing linear functionals subject to monotonicity and majorization constraints. To this end, define the function $H_G : [0,1] \to \mathbb{R}$ such that $H_G(G(\theta)) = \int_0^\theta \frac{1}{h_G(\kappa)} dG(\kappa)$. Let $\overline{H_G}$ denote the convex hull of H_G, i.e., the largest convex function that lies below H_G. By Proposition 2 of [Kleiner et al., 2020], X^* is an optimal solution to the above problem if the following holds: there exists a family I of disjoint intervals $\{[\alpha_i, \beta_i)\}_{i \in I}$ such that (i) $X^*(\theta) = X_{\text{PAM}}(\theta)$ if θ belongs to no interval in I and $X(\theta) = \frac{\int_{\alpha_i}^{\beta_i} X_{\text{PAM}}(\kappa)dG(\kappa)}{G(\beta_i)-G(\alpha_i)}$ when θ belongs to an interval $[\alpha_i, \beta_i)$, and (ii) $\overline{H_G}$ is affine over $[\alpha_i, \beta_i)$ for all $i \in I$ and $\overline{H_G} = H_G$ otherwise.

A function $z : \Theta \to \mathbb{R}$ is *convex-concave* if there is $\tilde{\theta} \in \Theta$ such that $z(\theta)$ is convex for $\theta \leq \tilde{\theta}$ and concave for $\theta \geq \tilde{\theta}$. Similarly, $z(\theta)$ is *concave-convex* if there is $\tilde{\theta} \in \Theta$ such that z is concave for $\theta \leq \tilde{\theta}$ and convex over $\theta \geq \tilde{\theta}$.

From the fact that the function $\frac{1}{h_G}$ is single-peaked (single-dipped) it follows that H_G is convex-concave (concave-convex). When H_G is convex-concave, then there is an optimal solution X^* such that

$$X^*(\theta) = \begin{cases} X_{\text{PAM}}(\theta), & \theta < \theta^*, \\ \frac{\int_{\theta^*}^{\bar{\theta}} X_{\text{PAM}}(\kappa)dG(\kappa)}{1-G(\theta^*)} & \theta \geq \theta^*, \end{cases}$$

for some $\theta^* \in \Theta$. This can be seen from Figure 4. Following our analysis for Theorem 4.2, one can see that this interim allocation rule corresponds to the interim allocation rule of a monotone disjoint queue mechanism that pools only high types (Figure 4).

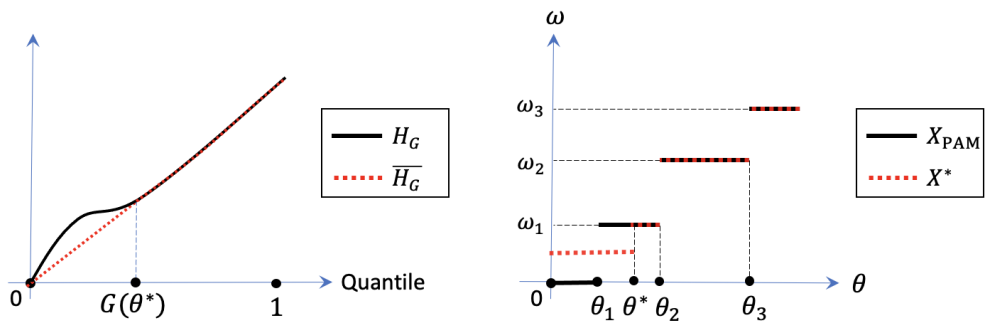

Figure 5: **Left panel:** $\overline{H_G}$ is affine to the left of $G(\theta^*)$ but not to its right. **Right panel:** the optimal rule $X^*(\theta)$. The corresponding monotone disjoint queue mechanism sends to q_1 objects of type ω_0, ω_1 respectively at rates $G(\theta_1), G(\theta^*) - G(\theta_1)$, objects of type ω_1 to q_2 at rate $G(\theta_2) - G(\theta^*)$, objects of type ω_2 to q_3 at rate $F(\omega_2)$, and objects of type ω_3 to q_4 at rate $F(\omega_3)$.

Similarly, when H_G is concave-convex, then there is an optimal solution X^* such that

$$X^*(\theta) = \begin{cases} \frac{\int_0^{\theta^*} X_{\text{PAM}}(\kappa) dG(\kappa)}{G(\theta^*)}, & \theta < \theta^*, \\ X_{\text{PAM}}(\theta) & \theta \geq \theta^*. \end{cases}$$

This can be seen from Figure 5. This interim allocation rule corresponds to an interim allocation rule that pools only low types, as demonstrated in that figure.

Construction of the optimal mechanism in the separable case. Constructing the optimal monotone disjoint queue mechanism in the separable case is generally similar to Figures 4 and 5. Consider the family I of disjoint intervals $\{[\alpha_i, \beta_i)\}_{i \in I}$ described earlier that characterizes the optimal interim allocation rule X^*. There are two types of queues in the associated optimal mechanism. First, there is a queue p_i corresponding to each interval $[\alpha_i, \beta_i)$, to which, for every $\omega_j \in \Omega$, objects of type ω_j are sent at rate r_j, where r_j is the measure assigned by G to

$$\{\theta \in [\alpha_i, \beta_i) : X_{\text{PAM}}(\theta) = \omega_j\}.$$

Second, for every $\omega_l \in \Omega$ there is (at most) one queue q_l to which only objects of type ω_l are sent: let (c_l, d_l) denote a maximal interval, not intersecting any of the intervals in I, such that $X_{\text{PAM}}(\theta) = \omega_l$ for every $\theta \in (c_l, d_l)$. Then, objects of type ω_l are sent to q_l at a rate $G(d_l) - G(c_l)$. By construction, the set of queues $\{p_i\}_i \cup \{q_l\}_l$ corresponds to a monotone disjoint queue mechanism with interim allocation rule X^*

Finally, we include two examples that illustrate the construction of the optimal mechanism with full and partial pooling, respectively.

Example E.1 (Construction of optimal mechanism with full pooling)**.** We consider the example in Section 3. In this case the virtual payoff function of the uniform distribution is $H_G(\theta) = \frac{1-G(\theta)}{g(\theta)} = 1-\theta$, which is strictly decreasing (see the left plot in Figure 6). Thus, by our previous arguments, $\overline{H_G}$ is affine over $[0,1]$ so full pooling is optimal. This means that any agent with type $\theta \in [0,1]$ is allocated at random an object of type ω_2, ω_1, or ω_0 with probability $F(\omega_2) = 0.25$, $F(\omega_1) = 0.25$, and $1 - N = 0.5$, respectively (see the right plot in Figure 6). The expected waiting cost is zero.

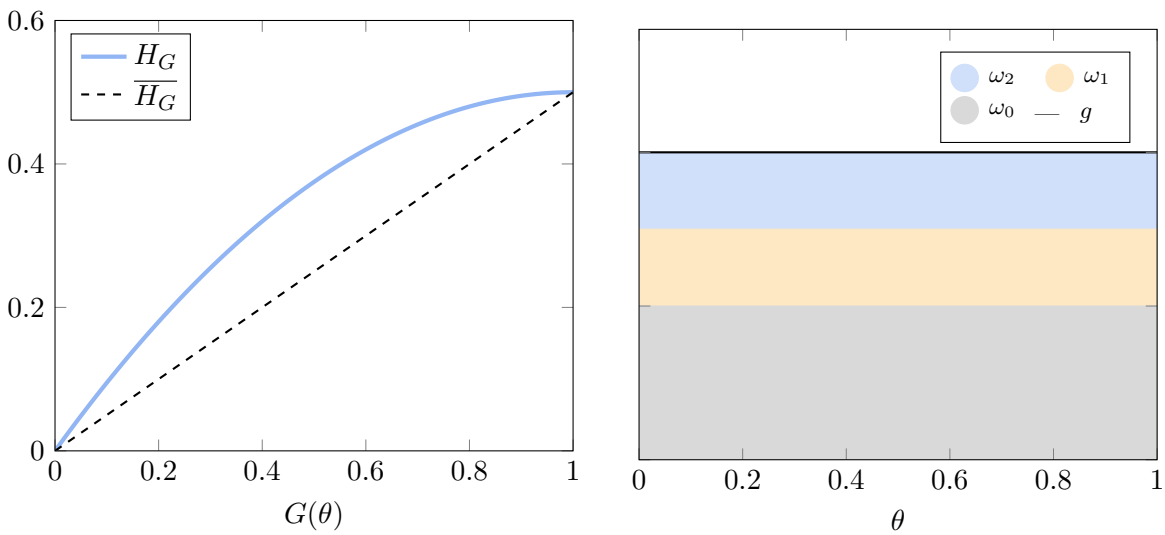

Figure 6: Construction of the optimal monotone disjoint queue mechanism in Example E.1.

F Proofs for Section 5

To prove Proposition 5.2, we prove the following stronger statement.

Proposition F.1. *The FCFS policy has a unique steady-state equilibrium. At this equilibrium, there are no two agents of distinct types θ, θ' such that $\theta > \theta'$ and the object assigned to the agent of type θ has a higher type than the object assigned to the agent of type θ'.*

Proof. For notational simplicity, let $\alpha_i = \omega_{n-i+1}$, for every positive integer $i \leq n+1$. Hence $\alpha_0 > \alpha_1 > \ldots > \alpha_n$. Let

$$t_1 = \sup_{t \in T}\{t : \exists \theta \in \Theta \text{ such that } a(t,\theta) > 0\}.$$

First, we will show that all objects of type α_1 are allocated at time t_1.

Claim F.2. *$M(t_1, \alpha_1) = F(\alpha_1)$, and $M(t_1, \alpha_i) = 0$ for all $i > 1$.*

Proof. The proof for $M(t_1, \alpha_1) = F(\alpha_1)$ is by contradiction. Suppose $M(t_1, \alpha_1) < F(\alpha_1)$. Then, by the left-continuity of $a(\cdot, \theta)$, there exists $\epsilon > 0$ such that, for any $t \in (t-\epsilon, t_1)$, $\int_{\theta \in \Theta} a(t_1, \theta) \, \mathrm{d}\, G(\theta) > 0$. On the other hand, by the regularity of $b(\cdot, \alpha_1)$, there exists some $\epsilon' > 0$ such that for all $t \in (t_1 - \epsilon', t_1)$, $b(t, \alpha_1) > 0$. Therefore, there exists a positive $\delta = \min\{\epsilon, \epsilon'\}$, such that for all $t \in (t_1 - \delta, t_1)$, $M(t, \alpha_1) > 0$. This implies that for any point $t \in (t_1 - \delta, t_1)$, there is a jump discontinuity in $b(t, \alpha_1)$. This contradicts the fact that $b(\cdot, \alpha_1)$ is monotone, since any monotone function has countably many jump discontinuities, by Froda's theorem.

Now we prove the claim for $i > 1$. The proof is by contradiction. Suppose $M(t_1, \alpha_i) > 0$ for some $i > 1$. Then, there must be $\theta \in \Theta$ such that $M(t_1, \theta, \alpha_i) > 0$. This contradicts the optimality of the decision rule $\mathcal{D}_\theta$ for agents of type θ; such agents can increase their payoff by switching to a new decision rule $\mathcal{D}'_\theta$ defined as follows.

$$\mathcal{D}'_\theta(t) = \begin{cases} \mathcal{D}_\theta(t) & \text{for } t \leq t_1 - \epsilon, \\ \alpha_1 & \text{for } t > t_1 - \epsilon, \end{cases}$$

where ϵ is a sufficiently small positive constant. □

Repeating the same argument as from the proof of Claim F.2 shows that there exists a nonnegative $t_2 < t_1$ such that $M(t_2, \alpha_2) = F(\alpha_2)$, and $M(t, \alpha_i) = 0$ for all $t > t_2$ and $i \geq 2$. Repeating this argument inductively shows that for every integer $j \in [2, n]$ there exists $t_j > 0$ such that $t_j < t_{j-1}$, $M(t_j, \alpha_j) = F(\alpha_j)$, and $M(t, \alpha_i) = 0$ for all $t > t_j$ and $i \geq j$.

Therefore, in any steady-state equilibrium of the FCFS policy, we must have a *positive assortative* steady-state assignment, i.e., there are no two agents of distinct types θ, θ' such that $\theta > \theta'$ and the object assigned to the agent of type θ has a lower type than the object assigned to the agent of type θ'. (Otherwise the agent of type θ would be better off if she chooses the cutoff decision rule of the other agent, due to the supermodularity of u.) This proves the latter claim of the proposition.

To prove the former claim (uniqueness), observe that a decision rule that sets $\mathcal{D}_\theta(t) = \alpha_j$ for all $t \in T$ guarantees an agent of type θ a payoff of $u(\theta, \alpha_j) - c(t_j)$ if $j \leq n$ and a payoff of $u(\theta, \omega_0)$ otherwise. We next use this fact to prove the uniqueness of the steady-state equilibrium.

Because the steady-state assignment is positive assortative, there exists some agent type $\widehat{\theta}$ such that all agents with type $\theta > \widehat{\theta}$ are assigned an object with type higher than ω_0, but all agents with type $\theta < \widehat{\theta}$ are not. Hence, the cutoff decision rule $\mathcal{D}_{\widehat{\theta}} \equiv \alpha_n$ should guarantee an agent of type $\widehat{\theta}$ a payoff of $u(\theta, \omega_0)$. This uniquely determines t_n, which in turn determines t_{n-1}, and so on. Hence, the steady-state equilibrium is unique. □

Proof of Proposition 5.3. Since $\mathcal{M}$ is positive assortative, then there exists $\{\theta_0, \ldots, \theta_k\}$, where $0 = \theta_0 \leq \theta_1 \leq \ldots \leq \theta_k = \overline{\theta}$, such that $X_{\mathcal{M}}(\theta') = X_{\mathcal{M}}(\theta'')$ holds for all $\theta', \theta'' \in (\theta_i, \theta_{i+1})$ and nonnegative integers $i < k$. Hence, $\mathcal{M}$ is regular. By the incentive compatibility of the mechanism and positive assortative assortative assignment, then, $V_{\mathcal{M}}(0) = 0$ must hold. To see why, suppose not. If

$V_{\mathcal{M}}(0) > 0$, then for any $\theta \in (0-\delta, 0)$, we must have $v_{\mathcal{M}}(\theta', \theta) > 0$, where $\delta > 0$ is a sufficiently small constant. But this would be a contradiction.

On the other hand, by Proposition F.1, FCFS has a unique steady-state equilibrium, namely $\mathcal{FCFS}$. Also, observe that the FCFS policy does not depend on the agents' type reports. Hence, it is incentive-compatible. Therefore, the steady-state equilibrium of FCFS has a corresponding reduced-form representation that satisfies the interim incentive compatibility constraints (A.1). By Proposition F.1, $\mathcal{FCFS}$ is positive assortative and regular; also, the payoff of agents of type 0 equals 0 at this equilibrium. Therefore, the conditions for the payoff-equivalence result of Theorem B.4 are met. By Theorem B.4, for every agent type θ, the payoff of an agent of type θ in $\mathcal{M}$ equals the payoff of an agent of type θ in $\mathcal{FCFS}$. □

Proof of Proposition 5.4. For any signal $\widehat{\omega} \in \widehat{\Omega}$, define

$$v(\widehat{\omega}) = \mathbb{E}_{\widehat{\omega} \sim \mu(\cdot|\omega), \omega \sim F_0} \left[\omega | \widehat{\omega}\right],$$

that is, $v(\widehat{\omega})$ gives the expected object type conditional on a signal realization $\widehat{\omega}$ being revealed to the agents. Consider $\{v(\widehat{\omega}) : \widehat{\omega} \in \widehat{\Omega}\}$ to be an *alternative* object space with $v(\widehat{\omega})$ corresponding to the object types. Because μ^* pools adjacent object types, then $v(\widehat{\omega}_1) \neq v(\widehat{\omega}_2)$ holds for all distinct $\widehat{\omega}_1, \widehat{\omega}_2 \in \widehat{\Omega}$. Therefore, by Proposition 5.2, the FCFS policy has a unique steady-state equilibrium when run with the alternative object types. This proves the claim. □

Proof of Theorem 5.5. We start by constructing the information disclosure policy μ^*, which consists of a finite signal realization space $\widehat{\Omega}$ and a family of distributions $\{\mu^*(\cdot|\omega)\}_{\omega \in \Omega}$ over $\widehat{\Omega}$. By Theorem 4.2, the optimal mechanism is a monotone disjoint queue mechanism, namely $\mathcal{M}$. Let $q_0, q_1, \ldots, q_k$ denote the queues used in the mechanism, where q_i has a higher type than q_j iff $i > j$. (We recall the definition of "higher type" from Section 4.)

For each queue q_i with $i \geq 1$, $\widehat{\Omega}$ contains a signal realization s_i, i.e., $\widehat{\Omega} = \{s_1, \ldots, s_k\}$. To construct $\{\mu^*(\cdot|\omega)\}_{\omega \in \Omega}$, we first need a few definitions. For each object type ω and queue q_i with $i \geq 0$, let $m_{\omega,i}$ be the steady-state rate of objects of type ω that are assigned to the queue q_i under $\mathcal{M}$. Then, for every $\omega \in \Omega$, define $\mu^*(\cdot|\omega)$ to be the probability distribution that assigns probability mass $\frac{m_{\omega,i}}{\sum_{j=0}^{k} m_{\omega,j}}$ to the signal s_i, for every $s_i \in \widehat{\Omega}$.

For every $s_i \in \widehat{\Omega}$, define $\overline{\omega}_i$ to denote the expected type of an object conditional on the object's having a corresponding signal realization s_i, i.e.,

$$\overline{\omega}_i = \mathbb{E}_{s_i \sim \mu^*(\cdot|\omega), \omega \sim F_0} \left[\omega | s_i\right].$$

Because $\mathcal{M}$ is a monotone disjoint queue mechanism, μ^* pools adjacent object types. For the same reason, $\overline{\omega}_i > \overline{\omega}_j$ iff $i > j$. We say a signal realization s_i has a higher type than a signal realization s_j iff $i \geq j$.

By Proposition 5.4, the FCFS policy under μ^* has a unique steady-state equilibrium, namely $\mathcal{F}$. Hence, to prove the claim, it remains to show that the payoff of every agent in $\mathcal{M}$ equals her payoff in the FCFS policy under μ^*. To this end, we need the following definitions.

We say an agent is *assigned to a signal* s_i in $\mathcal{F}$ if that agent has accepted an object with a corresponding signal realization s_i. For notational convenience, we suppose that an unassigned agent is assigned an object of type ω_0 (which is in abundant supply and has a type lower than any other object type in Ω). Similarly, in the FCFS policy under μ^*, we say that an unassigned agent is assigned to a signal realization s_0. The signal realization s_0, by definition, has a *lower* type than any signal realization in $\widehat{\Omega}$.

We say that $\mathcal{F}$ is *positive assortative with respect to* μ^* if, in $\mathcal{F}$, there are no two agents a, a', respectively, of distinct types θ, θ' such that $\theta > \theta'$ and a' is assigned to a signal of a higher type than a is. By Proposition F.1, $\mathcal{F}$ is positive assortative with respect to μ^*.

Let $X_{\mathcal{F}}(\theta)$ denote the interim expected object type that is assigned to an agent of type θ in $\mathcal{F}$. Since $\mathcal{F}$ is positive assortative with respect to μ^*, then there exists $\{\theta_0, \ldots, \theta_{k+1}\}$, where $0 = \theta_0 \leq \theta_1 \leq \ldots \leq \theta_{k+1} = \overline{\theta}$, such that the following properties hold:

(P1) For every nonnegative integer $i \leq k$, $X_{\mathcal{F}}(\theta) = X_{\mathcal{F}}(\theta')$ holds for all $\theta, \theta' \in (\theta_i, \theta_{i+1})$.

(P2) For every positive integer $i \leq k$, $G(\theta_{i+1}) - G(\theta_i) = F(\omega_i)$.

(P3) $X_{\mathcal{F}}(\theta) = 0$ if $\theta < \theta_1$ and $X_{\mathcal{F}}(\theta) > 0$ if $\theta > \theta_1$.

Similarly, since $\mathcal{M}$ is a monotone disjoint queue mechanism, then there exists $\{\theta'_0, \ldots, \theta'_k\}$, where $0 = \theta'_0 \leq \theta'_1 \leq \ldots \leq \theta'_{k+1} = \overline{\theta}$, such that the following properties hold:

(Q1) For every nonnegative integer $i \leq k$, $X_{\mathcal{M}}(\theta) = X_{\mathcal{M}}(\theta')$ holds for all $\theta, \theta' \in (\theta'_i, \theta'_{i+1})$.

(Q2) For every positive integer $i \leq k$, $G(\theta'_{i+1}) - G(\theta'_i) = F(\omega_i)$.

(Q3) $X_{\mathcal{M}}(\theta) = 0$ if $\theta < \theta'_1$ and $X_{\mathcal{M}}(\theta) > 0$ if $\theta > \theta'_1$.

Properties P1-P3 and Q1-Q3 imply that $\theta_i = \theta'_i$ for every nonnegative integer $i \leq k+1$. Therefore, the payoff-equivalence result of Theorem B.4 applies, which implies that, for any agent type θ, the payoff for every agent of type θ in $\mathcal{M}$ is equal to the payoff for every agent of type θ in the FCFS policy under μ^*. □

F.1 Example: FCFS policy with full information can be suboptimal

Example F.3. Suppose that $\Theta = [0, \bar{\theta})$, $\Omega = \{\omega_1, \omega_2\}$, $N = 1$, and let F_0 be the uniform distribution over Ω. Denote the median of distribution G by m.

Let μ denote the fully revealing information disclosure policy. Also, let μ' be an information disclosure policy with a singleton signal space, i.e., μ' discloses no information to agents. By

Proposition 5.2, the FCFS policy has unique steady-state equilibria under μ, μ', which we denote by $\mathcal{E}, \mathcal{E}'$, respectively. We next compute the social welfare in $\mathcal{E}, \mathcal{E}'$.

By Proposition F.1, $\mathcal{E}$ is positive assortative. Furthermore, from the proof of Proposition F.1 it follows that, for every object type, all agents that are assigned to an object of that type have the same waiting cost. Let t_1, t_2 denote the waiting costs for objects of type ω_1, ω_2, respectively. We note that, in $\mathcal{E}$, an agent of type m must be indifferent between accepting objects of type ω_1 and objects of type ω_2. Thus,

$$u(m, \omega_2) - t_2 = u(m, \omega_1).$$

It follows that $t_2 = m(\omega_2 - \omega_1)$. Consequently, every agent of type $\theta > m$ accepts only objects of type ω_2, and therefore has a payoff $V_{\mathcal{E}}(\theta) = \theta\omega_2 - m(\omega_2 - \omega_1)$. On the other hand, every agent of type $\theta < m$ accepts only objects of type ω_1. Since an agent of type $0 \in \Theta$ is indifferent between being assigned to an object and the outside option, then $t_1 = 0$. Hence, every agent of type $\theta < m$ has a payoff $V_{\mathcal{E}}(\theta) = \theta\omega_1$. Therefore, the social welfare in $\mathcal{E}$ is

$$W_{\mathcal{E}} = \frac{1}{2}\left(\mathbb{E}_\theta\left[\theta \mid \theta > m\right]\omega_2 - m(\omega_2 - \omega_1)\right) + \frac{1}{2}\mathbb{E}_\theta\left[\theta \mid \theta < m\right]\omega_1.$$

To compute the social welfare in $\mathcal{E}'$, we observe that in $\mathcal{E}'$ all agents have the same payoff $u(\theta, \omega_2)F(\omega_2) + u(\theta, \omega_1)F(\omega_1) = \theta(\omega_2 + \omega_1)/2$. The social welfare in $\mathcal{E}'$ then is

$$W_{\mathcal{E}'} = \mathbb{E}_{\theta \sim G}\left[\theta\right](\omega_1 + \omega_2)/2.$$

We observe that when, e.g., G is the uniform distribution over the unit interval,

$$W_{\mathcal{E}} = \frac{3\omega_1 + \omega_2}{8} < \frac{\omega_1 + \omega_2}{4} = W_{\mathcal{E}'},$$

which shows that social welfare under the full information disclosure policy μ is lower than under the no-information-disclosure policy μ'.

G Proofs for Section 6

G.1 Proof of Proposition 6.1

Throughout the proof, we refer to the setup of Section 2 by the *continuum model*, and the discrete setup of Section 6 by the *discrete model*. Let $\mathcal{H}_s$ denote the history of the process from time 0 until time $s \geq 0$. For a mechanism $\mathcal{M}$ in discrete model let the random variable $A_{\mathcal{M}}^{\mathcal{H}(s)}(\theta)$ denote the allocation timeline that the mechanism promises to an agent who arrives at time s and has type θ. (We note that $A_{\mathcal{M}}^{\mathcal{H}(s)}(\theta)$ can depend on the history $\mathcal{H}(s)$, and thus is a random variable.) Let $X_{\mathcal{M}}^{\mathcal{H}(s)}(\theta)$ denote the expected object type that is assigned to agent according to the allocation timeline $A_{\mathcal{M}}^{\mathcal{H}(s)}(\theta)$. For every s, we let $X_{\mathcal{M}}^{s}(\theta)$ be the random variable that takes the value $X_{\mathcal{M}}^{\mathcal{H}(s)}(\theta)$

when the history at time s is $\mathcal{H}(s)$.

Definition G.1. *A mechanism $\mathcal{M}$ is a* steady-state *mechanism if (i) for every $\theta \in \Theta$ there exists a CDF $Y_{\mathcal{M}}(\theta)$ defined over $[\omega_0, \omega_n]$ such that $X^s_{\mathcal{M}}(\theta)$ converges in distribution to $Y_{\mathcal{M}}(\theta)$ as s approaches infinity; when this holds we denote the mean of $Y_{\mathcal{M}}(\theta)$ by $X_{\mathcal{M}}(\theta)$ and refer to $X_{\mathcal{M}}(\cdot)$ as the* interim allocation rule *of $\mathcal{M}$, and (ii) $X_{\mathcal{M}}(\cdot) \in L^1(\Theta)$.*

We also sometimes alternatively say that $\mathcal{M}$ has a steady-state when it is a steady state mechanism.

As in the Proof of Theorem 4.2, let X_{PAM} denote the function $X_{\text{PAM}}(\theta) = F^{-1}(G(\theta))$ for $\theta \geq G^{-1}(1-N)$ and $X_{\text{PAM}}(\theta) = \omega_0$ for $\theta < G^{-1}(1-N)$. Define $\mathcal{F}$ to be the set of all increasing functions $X : \Theta \to \mathbb{R}_+$ such that X_{PAM} weakly majorizes X.

Lemma G.2. *For a steady-state mechanism $\mathcal{M}$, $X_{\mathcal{M}} \in \mathcal{F}$.*

Proof. The fact that $X_{\mathcal{M}}$ is increasing follows from the fact that ex post incentive compatibility implies interim incentive compatibility. The fact that X_{PAM} weakly majorizes $X_{\mathcal{M}}$ follows from the same argument as in the proof of Lemma D.4. □

Proof of Proposition 6.1. For every $X \in \mathcal{F}$, define

$$\mathcal{W}(X) = \int_0^{\bar{\theta}} X(\theta)\psi(\theta)\,\mathrm{d}G(\theta).$$

Let $X^* = \arg\max_{X \in \mathcal{F}} \mathcal{W}(X)$. Note that such X^* exists since $\mathcal{F}$ is closed. Since $\mathcal{F}$ is convex and $\mathcal{W}(\cdot)$ is linear in X, then by Bauer's Maximum Principle, X^* is an extreme point of $\mathcal{F}$.

We showed in the proof of Theorem 4.2 that each extreme point of $\mathcal{F}$ corresponds to disjoint queue mechanisms, and that for an extreme point X, $\mathcal{W}(X)$ gives the social welfare under the mechanism corresponding to X. Thus, the extreme point X^* corresponds to the optimal mechanism in the continuum model (which is a disjoint queue mechanism). Denote this mechanism by $\mathcal{M}^*$. Suppose that there are k disjoint queues in $\mathcal{M}^*$, and let $f_{\omega,i}$ denote the fraction of objects of type ω that are sent to queue i. Let W^* denote the social welfare under $\mathcal{M}^*$. We note that, for every market size λ, $W^*(\lambda) \leq W^*$, since the interim allocation rule of every mechanism in the discrete model belongs to $\mathcal{F}$.

In the discrete model, for any market size λ, let $\tilde{\mathcal{M}}(\lambda)$ denote the disjoint queue mechanism in which every object of type ω, upon its arrival, is sent to one of the queues $1, \ldots, k$ independently, such that the object is sent to queue i with probability $f_{\omega,i}$. From the concentration bounds of [Ashlagi et al., 2020] for the discrete model (their Section 5) it directly follows that (i) $\tilde{\mathcal{M}}(\lambda)$ has a steady state and, as λ approaches infinity, (ii) the steady-state (i.e., time-average) length of queue i under $\tilde{\mathcal{M}}(\lambda)$ in the discrete model approaches the length of queue i under $\mathcal{M}^*$ in the continuum

model, and (iii) for all but a measure zero set of agent types $\theta \in \Theta$,[23] the average object type assigned to an agent of type θ under $\tilde{\mathcal{M}}(\lambda)$ approaches the average object type assigned to an agent of type θ under $\mathcal{M}^*$.

By (ii) and (iii), for all but a measure zero set of agent types $\theta \in \Theta$, the steady-state payoff of an agent of type θ under $\tilde{\mathcal{M}}(\lambda)$ approaches the steady state payoff of an agent of type θ under $\mathcal{M}^*$, as λ approaches infinity. Therefore, $\lim_{\lambda\to\infty} \tilde{W}(\lambda) = W^*$. This fact, together with $\tilde{W}(\lambda) \leq W^*(\lambda)$ and $W^*(\lambda) \leq W^*$ (which holds due to the interim incentive-compatibility and individual rationality constraints), implies that $\lim_{\lambda\to\infty} \tilde{W}(\lambda) = \lim_{\lambda\to\infty} W^*(\lambda)$. □

G.2 FCFS in discrete markets

We will analyze the FCFS policy under μ^* in the discrete model of Section 6.1. To this end, we first need to define some notation related to the associated continuum model (where the Poisson arrival processes are replaced with flow arrivals of the same constant rate). Consider the unique steady state of the FCFS policy under μ^* in the continuum model. By the proof of Theorem 5.5, we can also consider this outcome to be the outcome of a disjoint queue mechanism $\mathcal{M}^*$, as defined next.

Without loss of generality, suppose that $\widehat{\Omega} = \{1, \ldots, k\}$ for $k > 0$. There is a queue in $\mathcal{M}^*$ associated with every element of $\widehat{\Omega}$. We also consider an additional auxiliary queue in $\mathcal{M}^*$ for agents not joining the pool. Thus, let $q_0, q_1, \ldots, q_k$ denote the queues in $\mathcal{M}^*$, and recall that q_i has a higher type than q_j if $i > j$. Define $Q = \{1, \ldots, k\}$. Let $\theta_0 = 0, \theta_1, \ldots, \theta_k, \theta_{k+1} = \overline{\theta}$ be such that an agent who reports a type $\theta \in [\theta_i, \theta_{i+1})$ for $i \geq 0$ is assigned upon arrival to the queue q_i. With slight abuse of notation, when the type θ of the agent is clearly known from the context, we also use q_θ to denote the queue that the agent joins. Let c_{q_i} denote the waiting cost incurred by an agent from waiting in queue q_i under the mechanism $\mathcal{M}^*$, and let $\overline{\omega}_i$ denote the expected object type that is sent to q_i. We also denote c_{q_i} by c_i when q_i is clearly known from the context.

Let the information disclosure policy μ^* of Theorem 5.5 consist of the finite signal realization space $\widehat{\Omega}$ and the family of distributions $\{\mu^*(\cdot|\omega)\}_{\omega\in\Omega}$ over $\widehat{\Omega}$. An *action* is a *threshold* belonging to $\widehat{\Omega} \cup \{\infty\}$ chosen by an agent which determines the lowest signal realization (drawn from μ^* and associated with an object) that the agent accepts if it is offered at the time of she takes the action. This threshold is the smallest (expected) object type that the agent accepts if offered. By convention, the action ∞ corresponds to the agent leaving the market immediately after taking the action. We let $A = \widehat{\Omega} \cup \{\infty\}$ denote the set of actions.

We now turn to analyzing the discrete model. In the proceeding analysis, by the first agent in the queue we mean the agent who is ahead of every other agent in the queue. The second agent in the queue is the agent who is ahead of every other agent but the first agent, and so on.

[23] The measure zero set is the set of agent types that are *indifferent* between joining at least two of the queues under the optimal monotone disjoint queue mechanism in the continuum model.

A 1-*strategy* is an element of A. The set of all 1-strategies is denoted by S_1. A 2-*strategy* is a tuple (s_2, s_1) where $s_2 : S_1 \to A$ and $s_1 \in S_1$. The set of all 2-*strategies* is denoted by S_2. Similarly, for every integer $i > 2$, an i-*strategy* is a tuple $(s_i, s_{i-1}, \ldots, s_1)$, where $s_i : \prod_{j=1}^{i-1} S_j \to A$ and $s_k \in S_k$ for every $k \leq i-1$. The set of all i-strategies is denoted by S_i. An element of $\cup_{i=1}^{\infty} S_i$ is called a *strategy*.

Every agent, upon arrival, adopts a strategy. In particular, when the queue length is i upon the arrival of an agent a, she adopts an i-strategy. Adopting an i-strategy $(s_i, s_{i-1}, \ldots, s_1)$ means that, while the agent remains at position i, she would take action $s_i(s'_{i-1}, s'_{i-2}, \ldots, s'_1)$ where s'_j is the j-strategy adopted by the agent at position j, for every $j \leq i-1$. If an agent ahead of agent a accepts an offer, the queue length is reduced by one; agent a moves to position $i-1$, and she adopts the $(i-1)$-strategy $(s_{i-1}, \ldots, s_1)$.

A *queue composition* is a finite sequence of an arbitrary length, namely $s = \langle s_k, \ldots, s_1 \rangle$, where $s_i \in S_i$ for every $i \leq k$. This queue composition corresponds to a queue where the i-th agent in the queue has adopted strategy s_i. We say an agent *faces a queue composition* $s = \langle s_k, \ldots s_1 \rangle$ if, upon her arrival, there are k agents waiting in the queue (excluding herself) such that the i-th agent in the queue has adopted strategy s_i, for every i.

A *strategy map* is a function T that assigns a $(k+1)$-strategy $T_{\theta,q}$ to every agent of type θ, where q is the queue composition that the agent faces upon her arrival and k denotes the length of q.

The Markov process $\mathcal{P}_T$ associated with a strategy map T is defined as follows. The state space of the process $\mathcal{P}_T$ is the set of all possible queue compositions. The state of the process $\mathcal{P}_T$ at a time is the queue composition at that time. The process $\mathcal{P}_T$ starts from an empty queue and evolves according to the following rules: Agents and objects arrive as specified in the discrete type model of Section 6. When an agent a arrives, she immediately enters the queue and adopts a strategy T_{θ_a, q_a}, where θ_a, q_a respectively denote the agent's type and the queue composition that she faces upon arrival. Objects are offered to agents in a first-come, first-served manner; so, every arriving object is first offered to the agent at the head of the queue; if the offer is declined, the object is offered to the next agent in the queue, and so on. Every agent accepts or declines offers according to her adopted strategy.

We say a strategy map *induces a steady state* if the Markov process $\mathcal{P}_T$ has a unique steady state.

Given a strategy map T, an agent can assign a payoff to any strategy s that she may choose upon her arrival (taking the decisions of other agents as given by the strategy map T). Then, we say the agent's *ex ante payoff* from choosing strategy s equals $u(\theta_a, s_\omega) - s_c$, where s_ω is the expected object type she allocates and s_c is the expected waiting cost she incurs under strategy s. When the strategy map T is clearly known from the context, we suppress it from the notation, as in the previous sentence.

Definition G.3. *An ϵ-equilibrium is a strategy map T for which the following holds:*

i. T induces a steady state.

ii. $\lim_{\lambda\to\infty} \mathcal{D}_T(\lambda) = 0$, where $\mathcal{D}_T(\lambda)$ is the measure of agents, at the steady state induced by T, who can increase their ex ante payoff by choosing a different strategy upon arrival than the one assigned to them by the strategy map T.

Let $i > 0$ be an arbitrary integer. A strategy $(s_i, s_{i-1}, \ldots, s_1)$ is called a *constant strategy* if there exists $\tau \in A$ such that $s_j(\cdot) \equiv \tau$ for every $j \leq i$. More specifically, such a strategy is also called a *constant strategy* τ.

Definition G.4 (The strategy map T^*)**.** *The strategy map T^* assigns to every agent a constant strategy as follows. Consider an arbitrary agent a, and suppose that she faces a queue composition q upon her arrival. Let $u^q_{a,\tau}$ denote the ex ante payoff of the agent, computed at her arrival time, from choosing a constant strategy $\tau \in A$. Then, T^* assigns the agent the constant strategy $\tau^* = \arg\max_{\tau\in A} u^q_{a,\tau}$.*

Conjecture G.5 (Concentration bound)**.** *T^* induces a steady state which satisfies the following condition. Consider an agent a arriving at the steady state and suppose that the agent is of type θ. Let the random variable $c^\lambda_a(\tau)$ denote the waiting cost that the agent incurs if she chooses a constant strategy τ. For every $\kappa > 0$ there exists λ_κ such that, if $\lambda > \lambda_\kappa$, then for an arriving agent a,*

$$\mathbb{P}\left[(1-\kappa)c_{q_\theta} \leq c^\lambda_a(\tau) \leq (1+\kappa)c_{q_\theta}\right] > 1-\kappa. \tag{G.1}$$

Simulations supporting the conjecture. We next provide simulation results that support the above conjecture. As in Section 6.2, we assume that G is the uniform distribution and that $c(t) = \gamma t$ with $\gamma = 0.05$, and then vary λ, the parameter that scales the agents' and objects' arrival rates. For our simulation we consider the case of $\widehat{\Omega} = \{1, 2\}$, i.e., the agents may observe two possible signal realizations, the *high* realization 2 and the *low* realization 1. Let $\overline{\omega}_i$ denote the expected object quality conditional on the signal realization i. Suppose that (objects with) signal realization i arrive according to a Poisson process with rate λ.

Agents arrive according to a Poisson process with rate 3λ. They wait in a FCFS queue and take actions (i.e., decide to accept or decline offers) according to the strategy map T^* defined above. We run the process until 10^8 agents arrive. Then, for every arriving agent and every $\tau \in \widehat{\Omega}$ we compute $c^\lambda_a(\tau)$, the agent's waiting cost if she adopts a constant strategy τ upon her arrival (taking the other agents' strategies as given in T^*). Fixing λ, τ, this gives an empirical distribution D^λ_τ containing 10^8 data points for the agents' waiting costs. Finally, for every arrival rate λ, we compute an interval $[c^l_\tau, c^u_\tau]$ for each $\tau \in \widehat{\Omega}$. The interval is chosen so that it contains 95% of the agents' waiting costs

$c_a^\lambda(\tau)$. More precisely, the interval excludes 2.5% of the data points from each tail of the empirical distribution D_τ^λ.

Figure 7 plots c_τ^l, c_τ^u for $\tau = 1, 2$ while varying λ. The figure also marks the values $c_1 = c_{q_1} = 1$ and $c_2 = c_{q_2} = \frac{1}{3}$ (the waiting costs associated with queues q_1 and q_2 in the continuum model). We observe that the length of the $[c_\tau^l, c_\tau^u]$ approaches 0 as λ grows large, suggesting that the discussed concentration bound holds.

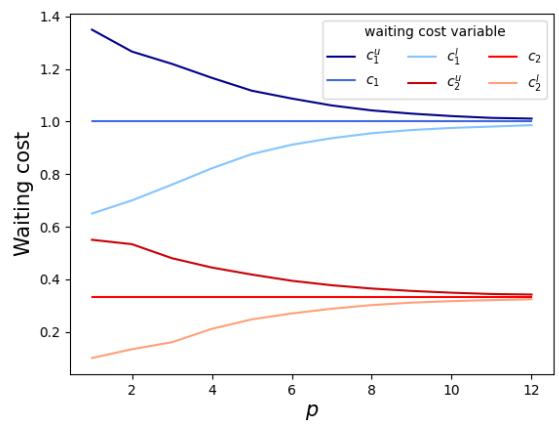

Figure 7: We vary $\lambda = 10 \times 2^p$ and plot c_i, c_i^l, c_i^u for $i = 1, 2$.

□

Proposition G.6. *If Conjecture G.5 holds, then for every $\epsilon > 0$, there exists $\lambda_\epsilon > 0$ such that, when $\lambda > \lambda_\epsilon$, T^* is an ϵ-equilibrium.*

Proof. Define

$$\Theta_\delta = \Theta - \{\theta \in [\theta_i - \delta, \theta_i + \delta] : i \in Q\}.$$

Also, let G_δ denote the G-measure of Θ_δ. We note that

$$\lim_{\delta \to 0} G_\delta = 1. \tag{G.2}$$

Claim G.7. *For every $\delta > 0$, there exists $\epsilon_\delta > 0$ such that for every $\theta \in \Theta_\delta$ and every queue $q_i \neq q_\theta$,*

$$\frac{u(\theta, \overline{\omega}_{q_\theta}) - c_{q_\theta}}{u(\theta, \overline{\omega}_i) - c_i} > 1 + \epsilon_\delta.$$

Proof. Consider an agent of type $\theta_j \in \{\theta_1, \ldots, \theta_k\}$. Such an agent is indifferent between joining queues q_j and q_{j-1}, i.e., $u(\theta_j, \overline{\omega}_j) - c_j = u(\theta_j, \overline{\omega}_{j-1}) - c_{j-1}$. Supermodularity of u, together with the indifference conditions, implies that for every agent of type $\theta \in (\theta_j + \delta, \theta_{j+1} - \delta)$, there exists

a number $\Delta_j > 0$ such that the agent's payoff is reduced by at least Δ_j if she joins a queue other than q_j. In particular,

$$\Delta_j = (u(\theta_j + \delta, \overline{\omega}_j) - u(\theta_j + \delta, \overline{\omega}_{j-1})) - (u(\theta_j, \overline{\omega}_j) - u(\theta_j, \overline{\omega}_{j-1})),$$

which is a lower bound on how much payoff the agent loses if she joins a queue with a type lower than j. Noting that $q_\theta = j$, we can then write the following for any i with $q_i \neq q(\theta)$

$$\frac{u(\theta, \overline{\omega}_{q_\theta}) - c_{q_\theta}}{u(\theta, \overline{\omega}_i) - c_i} > 1 + \frac{\Delta_j}{u(\theta, \overline{\omega}_i) - c_i}.$$

Defining ϵ_δ to be the minimum over $i, j \in \{1, \ldots, k\}$ with $i \neq j$ of the right-hand side of the above inequality concludes the proof. □

We are now ready to prove the proposition. Consider the FCFS policy under μ^* in the discrete model. Fix an arbitrary $\delta > 0$, and consider an agent a of type $\theta \in \Theta_\delta$ upon arrival. From Claim G.7, it follows that there exists arbitrarily small $\kappa_\delta > 0$ such that for every queue $q_i \neq q_\theta$,

$$\frac{u(\theta, \overline{\omega}_{q_\theta}) - (1 + \kappa_\delta)c_{q_\theta}}{u(\theta, \overline{\omega}_i) - (1 - \kappa_\delta)c_i} > 1 + \kappa_\delta/2 > 1. \tag{G.3}$$

Let $\tau \in A$ denote the constant strategy assigned to a by T^*. Define event $\mathfrak{e}$ as

$$(1 - \kappa_\delta)c_{q_\theta} \leq c_a^\lambda(\tau) \leq (1 + \kappa_\delta)c_{q_\theta}.$$

When event $\mathfrak{e}$ holds, (G.3) implies that an agent of type belonging to Θ_δ cannot increase her payoff by deviating to another strategy than the one assigned her by T^*. If Conjecture G.5 holds then there exists $\lambda_{\kappa_\delta} > 0$ such that $\mathbb{P}[\mathfrak{e}] \geq 1 - \kappa_\delta$ when $\lambda > \lambda_{\kappa_\delta}$. This means that at least a measure $(1 - \kappa_\delta)G_\delta$ of the agents would not be able to increase their payoff by deviating to another strategy than the one assigned them by T^*. By (G.2), we can choose δ sufficiently small such that $1 - (1 - \kappa_\delta)G_\delta \leq \epsilon$. This concludes the proof. □

H Separable utilities and heterogeneous default options

We next consider an application of our main result to a setting where the agents' utility functions are not separable due to the presence of their heterogeneous *default options.* An agent of type θ has a default option of quality $\beta(\theta)$; the planner may assign agents to their default options at any time (without facing capacity constraints such as those for objects). Conditional on not being assigned to a default option, an agent's utility is separable, i.e., it is defined by the function $u(\theta, \omega) = \theta\omega$ as usual.

We suppose that $\beta(\theta)$ is continuous and increasing in θ, and that every agent can only *deflate* her type; i.e., she can pretend to be of a lower type than her actual type, but not vice versa.

Mechanisms are defined as in our main setup, with the individual rationality and incentive compatibility constraints adapted to incorporate the default options. For a mechanism with interim allocation and payment rules $X, t : \Theta \to \mathbb{R}_+$, the individual rationality constraint is $u(\theta, X(\theta)) - t(\theta) \geq \theta\beta(\theta)$, and the incentive compatibility constraint is

$$\theta \max(\beta(\theta), X(\theta)) - t(\theta)) \geq \theta \max(\beta(\theta'), X(\theta')) - t(\theta')).$$

In particular, it is assumed that if the planner assigns an agent to a default option, it would be the default option corresponding to the agent's type report. (This can be interpreted as a type of commitment power that helps the planner impose incentive compatibility.)

All else remains the same as in our main setup. The objective is finding the optimal mechanism, i.e., the individually rational and incentive-compatible mechanism with the highest social welfare. We will show that our main analysis can be applied here and the optimal mechanism would be a monotone disjoint queue mechanism.

We first define a *transformed market* where the agent' utility function is defined by $v(\theta, \omega) = \theta \max(0, X(\theta) - \beta(\theta))$ and the agents have no default options, as in our main setup. All else in the transformed market is the same as in the original market. We will show that (i) the optimal mechanism $\mathcal{N}^*$ in the transformed market is a disjoint queue mechanism, and that (ii) the same mechanism remains optimal in the original market.

For step (i), we first note that v is a supermodular utility function. Thus, Theorem 1 of [Rochet, 1987] applies and allows us to characterize an incentive-compatible mechanism with a monotone interim allocation rule $X : \Theta \to \mathbb{R}_+$ and an interim payment rule $t : \Theta \to \mathbb{R}_+$. Further, by the proof of the theorem, the payment rule t associated with an allocation rule X must be

$$t(\theta) = v(\theta, X(\theta)) - \int_0^\theta \max(0, X(\tau) - \beta(\tau))\mathrm{d}\tau.$$

From substituting the payment formula in an agent's payoff function and integrating the result over all agent types, it follows that the social welfare under an incentive-compatible and individually rational mechanism with interim allocation rule X in the transformed problem equals

$$\int_0^{\bar{\theta}} \int_0^\theta \max(0, X(\tau) - \beta(\tau))\mathrm{d}\tau \mathrm{d}G(\theta),$$

which we denote by $W(X)$. The planner's problem in the transformed market thus can be written as

$$\sup_{X \in \mathcal{S}} \int_0^{\bar{\theta}} \int_0^\theta \max(0, X(\tau) - \beta(\tau))\mathrm{d}\tau \mathrm{d}G(\theta). \tag{H.1}$$

We observe that the functional $W(X)$ is convex in X. Thus, the Bauer's maximum principle (Theorem D.7) applies and implies that the solution to the planner's problem $X^* \in \mathcal{S}$ is an extreme point of $\mathcal{S}$. Recall that the extreme points of $\mathcal{S}$ are associated with disjoint queue mechanisms, as established in our main analysis in Section D. Let the interim allocation rule X^* denote the solution to the planner's problem in the transformed market, and denote its associated payment rule by t^*. Also, let $\mathcal{N}^*$ denote the corresponding disjoint queue mechanism.

In the last step, we show that the disjoint queue mechanism $\mathcal{N}^*$ is an optimal incentive-compatible and individually rational mechanism for the original market. First, we observe that the incentive compatibility constraint in the transformed market implies the incentive compatibility constraint in the original market: for every $\theta, \theta' \in \Theta$ with $\theta \geq \theta'$,

$$\begin{aligned}
&v(\theta, X^*(\theta)) - t^*(\theta) \geq v(\theta, X^*(\theta')) - t^*(\theta') \\
\Rightarrow\, &v(\theta, X^*(\theta)) + \theta\beta(\theta) - t^*(\theta) \geq v(\theta, X^*(\theta')) + \theta\beta(\theta') - t^*(\theta') \\
\Rightarrow\, &\theta \max(\beta(\theta), X(\theta)) - t(\theta) \geq \max(\beta(\theta'), X(\theta')) - t(\theta'),
\end{aligned}$$

which is the incentive compatibility constraint for the original problem.

Finally, we note that the social welfare under $\mathcal{N}^*$ in the original market is precisely $\mathbb{E}_{\theta\sim G}[\theta\beta(\theta)]$ higher than in the transformed market, because $X^*(\theta) < \beta(\theta)$ implies $X^*(\theta) = t^*(\theta) = 0$, by the optimality and individual rationality of $\mathcal{N}^*$. On the other hand, the social welfare under any incentive-compatible and individually rational mechanism $\mathcal{N}$ in the original market is at most $\mathbb{E}_{\theta\sim G}[\theta\beta(\theta)]$ higher than in the transformed market. This shows the optimality of $\mathcal{N}^*$ in the original market.